

Intervening Against the Fed*

Alexander Rodnyansky[†]

The University of Cambridge

Yannick Timmer[‡]

Federal Reserve Board

Naoki Yago[§]

The University of Reading

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Abstract

This paper studies the effectiveness and mechanism of foreign exchange interventions (FXIs) for mitigating US monetary policy spillovers. Without interventions, contractionary US monetary policy shocks trigger foreign exchange depreciations, raise risk premiums, and induce portfolio outflows, thereby reducing foreign stock prices. The stock prices of firms with US dollar debt decline significantly more, indicating a strong role for a balance sheet channel. However, “intervening against the Fed” stabilizes exchange rate depreciations, the risk premium, and portfolio flows. FXIs also disproportionately cushion the reduction in stock prices for firms with US dollar debt, muting the balance sheet channel of exchange rates.

Keywords: Foreign Exchange Interventions, Monetary Policy Spillovers, Balance Sheet Channel, Exchange Rates, Dollar Debt

JEL Classification Codes: E44, E52, F31, F32, F41

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[†]Faculty of Economics, The University of Cambridge, Austin Robinson Building, Sidgwick Avenue, Cambridge, CB3 9DD, United Kingdom.

[‡]Corresponding author. Monetary Affairs Division, Federal Reserve Board, 20th Street and Constitution Avenue, Northwest, Washington, DC 20551, United States. Email: yannick.timmer@frb.gov. +12026649457

[§]ICMA Centre, Henley Business School, The University of Reading, Whiteknights, Reading, RG6 6DL, United Kingdom.

1 Introduction

The Federal Reserve is an important driver of the “Global Financial Cycle” and changes in US monetary policy can have major spillovers to the global economy. When the Fed tightens monetary policy, the US dollar appreciates, and global stock prices as well as credit decline (Rey 2015, Miranda-Agrippino and Rey 2020, Gürkaynak et al. 2021). Although both academics and policymakers have signaled the first-order importance of how countries could protect themselves against this unexpected tightening of monetary policy, there is limited consensus on the effectiveness of different measures.

Foreign exchange interventions (FXIs) are an increasingly popular policy tool among central banks attempting to insulate themselves from those spillovers. For instance, on September 21, 2022, the Federal Open Market Committee (FOMC) announced to increase the Fed funds rate by 0.75 percentage points, after which the Japanese yen depreciated strongly. However, when the Bank of Japan intervened in the FX market by selling the US dollar and buying the Japanese yen, the depreciation immediately reversed (Figure 1). This example shows how “intervening against the Fed” can potentially offset US monetary spillovers to exchange rates. Despite their widespread use, systematic evidence of how FXIs function as a tool to insulate financial markets and the economy from US monetary policy spillovers is elusive. We aim to fill this gap in the literature.

We first provide evidence on how US monetary policy spills over if countries do not counter-intervene. When US monetary policy unexpectedly tightens, domestic exchange rates depreciate against the US dollar, the currency risk premium rises, and investors withdraw their portfolios from foreign countries. Furthermore, US monetary policy tightening leads to a fall in stock prices, which is disproportionately so for firms that borrow in US dollars. The depreciation of the domestic exchange rate reduces firms’ net worth due to higher debt repayment, leading to lower stock prices (Krugman 1999, Céspedes et al. 2004). Then we show that intervening against the Fed, by unexpectedly selling the US dollar in response to a contractionary monetary policy shock, mitigates exchange rate depreciation and stock price declines, but more so for firms with US dollar debt. When US monetary policy tightens and central banks counter-intervene, exchange rates and stock prices for firms with and without US dollar debt remain statistically and economically unchanged. These results suggest that intervening against the Fed mutes the balance sheet channel of exchange rates triggered by US monetary policy and can protect countries from exposure to the Global Financial Cycle. Moreover, intervening against the Fed successfully prevents a rise in the currency risk premium and portfolio equity outflows, further stabilizing financial markets over and above the balance sheet channel.

Recent theory suggests that FXIs can affect the exchange rate and improve welfare (Gabaix and Maggiori 2015, Cavallino 2019, Amador et al. 2020, Fanelli and Straub 2021, Maggiori 2022, Itskhoki and Mukhin 2023). However, empirical evidence on the effectiveness of FXIs is limited by identification challenges. Both US monetary policy and FXIs are endogenously

conducted, taking current and future information into account. Moreover, changes in monetary or FX policy are inextricably linked and hard to isolate from other factors. These problems make it harder to identify the channels through which US monetary policy spills over to other countries and whether FXIs are successful in mitigating them. We overcome these issues in several ways.

Our novel identification method employs a multi-event study high-frequency approach by combining daily data on FXIs with firm-level stock returns, exchange rates, as well as risk premia and portfolio flows around US monetary policy decisions.¹ In particular, we use an event-study local projection difference-in-differences (LP-DID) approach in the spirit of [Dube et al. \(2023\)](#) around each FOMC meeting. The advantages of using financial market data around FOMC events are twofold. First, compared to outcomes only available at lower frequencies, such as investment, they can be measured at a daily frequency, which mitigates the aggregation bias introduced by aggregating monetary policy shocks or FXIs to lower frequencies. Second, market prices are forward-looking and therefore respond more quickly as they incorporate expected future economic outcomes, such as slower-moving variables like firms' investments and profits, that would not appear in contemporaneous balance sheet measures. Third, our identification strategy helps to identify a counterfactual of what would have happened if countries had not intervened. Simply observing exchange rates and stock prices around an intervention without comparing them to a counterfactual can lead to misleading conclusions.

To measure the effect of interventions, we use daily FXI data from 13 countries. We define expected and unexpected counter-interventions as situations in which the central bank sells (buys) US dollars when the Fed unexpectedly tightens (loosens) policy. We call this policy “intervening against the Fed” as selling US dollars in exchange for domestic currency is intended to put appreciation pressure on the domestic exchange rate by absorbing its supply and vice versa. We then define intervening against the Fed if the central bank counter-intervenes within five days after the FOMC meeting. To address the potential concern that FXIs may be driven by economic fundamentals that could bias our estimates, we also estimate deviations from an FX counter-intervention policy rule incorporating observable and unobservable macroeconomic and financial characteristics, as well as historical responses to Federal Reserve policy. Deviations from the FXI policy rule can be interpreted as unexpected FXIs, helping to isolate the effects of potential confounding factors and estimate the direct effects of interventions.

For US monetary policy, we use cleanly identified high-frequency shocks by [Nakamura and Steinsson \(2018\)](#). The high-frequency monetary shocks are estimated by a change in Fed funds futures in a 30-minute time window around the FOMC announcement so that monetary shocks are orthogonal to the limited amount of information revealed in this narrow time window. Using this monetary surprise can be seen as an external driver of changes in the exchange rate,

¹We refer to daily data as high-frequency. For the market microstructure literature, this definition may be inappropriate, but as the international finance literature predominantly focuses on monthly or quarterly changes, we consider our approach high-frequency within that tradition.

allowing us to compare the responses of exchange rates and stock prices to FOMC announcements in countries that do and do not counter-intervene against the US dollar shortly after the announcement.

There are many channels through which the US monetary policy can spill over to other countries under integrated financial and goods markets. Higher US monetary policy rates, *ceteris paribus*, predict an appreciation of the US dollar on impact due to interest rate differentials in the uncovered interest parity (UIP) equation. In the financial market, this dollar appreciation lowers stock prices through a balance sheet channel by increasing debt repayments for firms that borrow in US dollars ([Gopinath and Stein 2021](#)). Even if firms do not have dollar debt, monetary policy can affect stock prices by increasing currency risk premia ([Lustig et al. 2011](#)) or triggering portfolio flows ([Converse and Mallucci 2023](#), [Converse et al. 2023](#)). In the goods market, according to an expenditure switching channel, the depreciation of domestic exchange rates relative to the US dollar would then increase exports and raise the stock prices of firms ([Mundell 1957](#), [Fleming 1962](#)). In addition, a tighter US monetary policy may lower demand and imports in the US, with adverse effects on foreign economies ([Gourinchas 2018](#)).

Our high-frequency approach, coupled with firm-level stock prices and information on the currency decomposition of corporate debt, allows us to compare the cross-sectional heterogeneity in firms' stock price responses within each country, at a given point in time, to a contractionary monetary policy shock, ruling out the possibility that other macroeconomic factors or other channels, such as the risk premium and portfolio rebalancing channel, are driving the response. We find that, when countries do not counter-intervene against the Fed, stock prices of firms with US dollar debt decline immediately after US monetary policy unexpectedly tightens. Quantitatively, a 10 basis point surprise increase in the federal funds rate is associated with a 3% decline in stock prices for firms with US dollar debt on the day of the FOMC meeting, building up to almost 10% after three days. In contrast, firms without US dollar debt see a decline in the stock price of less than 1%. This is explained by an increase in the US equity and currency risk premiums, as well as equity portfolio outflows, which affect both firms with and without dollar debt. The stark differential response, however, strongly suggests that US monetary policy affects stock prices abroad through a balance sheet channel.

To shed more light on this channel, we move toward the effect of US monetary policy on exchange rates. If the balance sheet channel of US monetary policy is at work, one would expect the behavior of exchange rates to mirror that of the stock price for firms with US dollar debt. And, indeed, for countries that do not counter-intervene, exchange rates depreciate strongly after the US monetary policy shock. Quantitatively, a 10 basis point increase in the US monetary policy rate leads to a persistent 2-3% depreciation of foreign currency. We contribute to the literature by providing supporting evidence in favor of the effectiveness of FXIs on both the stock market and exchange rates, making progress on the endogeneity issues discussed by [Gabaix and Maggiori \(2015\)](#) and [Maggiori \(2022\)](#).

The depreciation of the exchange rate may also trigger an expenditure switching channel,

increasing demand from foreigners due to lower prices, in which case exporting firms would benefit disproportionately (Mundell 1957, Fleming 1962). However, the contractionary US monetary policy shock reduces demand from the US via intertemporal substitution and therefore also demand for exports, potentially offsetting the positive effect from the exchange rate (Gourinchas 2018).² When countries do not intervene and the exchange rate depreciates, the stock prices of exporters remain stable while those of non-exporters fall, suggesting that the negative demand effects from the contractionary monetary policy shock and the positive effects from the expenditure switching channel offset each other.

Turning to the role of FXIs in US monetary policy spillover, we estimate the effect of “intervening against the Fed” on the stock prices across the firm distribution, as well as the exchange rate, currency risk premiums, and portfolio flows. By tracking the effect of the FXIs within a short window after the FOMC meeting, we likely capture the central bank’s decision to “intervene against the Fed”. If, instead, we studied the unconditional effects of FXIs on exchange rates, it would be more difficult to understand the causal effect of the interventions. For instance, it may be possible that the central bank intervenes because it receives a signal about the economy that is unobservable, making it more difficult to understand the counterfactual response of the exchange rate if the central bank had not intervened. As we employ an event study in a short window around the FOMC announcement, the market movements are likely driven by US monetary policy and, without the FOMC decision, would have remained systematically unchanged, as the pre-trends indicate, allowing us to estimate a more precise counterfactual of not intervening.

Using these unexpected monetary shocks and FXIs, we present evidence that “intervening against the Fed” mitigates market disturbances when US monetary policy changes suddenly. When the Fed hikes rates unexpectedly and the country sells the US dollar and buys domestic currency to counteract the surprise, the domestic currency does not depreciate against the US dollar, the currency risk premium and portfolio flows remain stable, and stock prices for both firms with and without US dollar debt remain unchanged. These results suggest that FXIs offset the spillover of US monetary policy via the balance sheet channel: it mitigates a depreciation of the exchange rate and consequently higher debt repayments for firms with US dollar debt.³ Moreover, FXIs are also successful in preventing a rise in the currency risk premium and portfolio outflows, further stabilizing overall markets and offsetting the spillover of US monetary policy through a financial market channel. Our results highlight the importance of taking FXIs into account when analyzing monetary policy spillovers. When we pool no-intervention and intervention events, it is difficult to find evidence in favor of US monetary

²Under dominant currency pricing (Gopinath et al. 2020), export quantities are not expected to increase in the short-run, but revenues in domestic currencies are, also with positive implications for the stock price of exporters.

³One potential concern could be that firms use derivatives to hedge their exchange rate risk. While we do not have derivative data available, recent studies suggest that only a small fraction of firms hedge their FX risk with derivatives; see, e.g., Casas et al. (2022). Moreover, we introduce a statistical approach that proxies whether firms hedge their exchange rate risk, and we do not find that those firms affect our results. See also discussion in Appendix A.

policy spillovers, as this pooling masks important heterogeneity in the effects of US monetary policy abroad.

To credibly counter-intervene against the Fed to stabilize their exchange rate, countries need to have accumulated a sufficient level of reserves. In fact, since the Asian financial crisis in the 1990s, emerging market economies have built up reserves to hedge against rollover risk (Bianchi et al. 2018).⁴ When reserve levels are low, central banks' attempts to stabilize the exchange rate and consequently the economy may be unsuccessful as they lack the credibility to do so in the future (Fanelli and Straub 2021). Empirically, we find evidence that countries that have a large share of FX reserves are more effectively counter-intervening against the Fed relative to their counterparts. This result suggests that the buildup of reserves over the last decades could have dampened the role of US monetary policy spillovers if countries had counter-intervened.

While FXIs may prevent a depreciation in response to a contractionary monetary policy shock, they do not mitigate the negative demand effects on imports originating from the US. The benefits of the contractionary monetary policy shock through the expenditure-switching channel for exporters may therefore vanish. The costs, however, through reduced demand, remain the same. Consequently, FXIs may harm exporters through a less depreciated exchange rate. Indeed, we find suggestive evidence that when countries do intervene, the effects of intervening against the Fed are negative for exporting firms, especially those without US dollar debt. This presents a tradeoff for the central bank. Intervening against the Fed helps stabilize risk premia, portfolio flows, and the stock prices of firms, especially those with US dollar debt—but it can harm exporters by appreciating the domestic currency. A large exporting country may therefore choose not to intervene, prioritizing export competitiveness, while countries with significant US dollar debt exposure may be more inclined to intervene.

Our identification strategy exploits the deviation from an estimated FXI policy rule and extracts the unexpected component of FXI in an attempt to use exogenous variation. However, this approach is not free of limitations, since it is difficult to find a perfect instrument in a cross-country panel setting, as discussed in Fratzscher et al. (2019). There can be other confounding factors that affect both our dependent variable (stock returns and the exchange rate) and our independent variable (FXI) concurrently and introduce an omitted variable bias to our estimates. It is still possible that unwarranted market disturbances or negative news about the economy that are not captured in the FXI rule may affect stock prices and the exchange rate in response to US monetary policy, while at the same time inducing the central bank to intervene in the FX market.

While we acknowledge this limitation and encourage the subsequent literature to further improve upon this identification challenge, it is unlikely that this omitted variable fully explains our results. The exchange rate tends to depreciate systematically in “bad times” against the

⁴For example, the Swiss National Bank has sold the franc to counter the overvaluation due to its safe-haven status and boost the export industries (Jordan 2017). Chinese foreign reserves increased from 733 billion US dollars in July 2005 to 3.99 trillion US dollars in June 2014 (Das 2019).

US dollar ([Hassan et al. 2023](#)). When the negative shock hits the economy and stock prices fall, one would expect the central bank to sell the US dollar to prevent the depreciation of the exchange rate. However, we find that selling the US dollar against the domestic currency is rather associated with an appreciation of the domestic currency and an increase in stock prices relative to the no-intervention case. This renders it unlikely that unobservable confounders are driving the relationship between FXIs and those outcome variables.

2 Literature Review

Our paper contributes to the FXI literature and to how US monetary policy spills over to other countries. Our novel event study approach exploits daily firm-level stock prices across several countries, as well as exchange rates combined with FXI data and US monetary surprises purely identified by a high-frequency approach, see, e.g., [Nakamura and Steinsson \(2018\)](#).

A large empirical literature studies the effect of FXIs on exchange rates ([Dominguez and Frankel 1993](#), [Dominguez 2003](#), [Dominguez et al. 2013](#), [Adler et al. 2019](#), [Fratzscher et al. 2019](#); [2023](#), [Hofmann et al. 2019](#), [Blanchard et al. 2015](#), [Kuersteiner et al. 2018](#), [Fatum and Hutchison 2010](#)).⁵ However, none of these papers study the effect of FXIs on stock prices across the firm distribution, nor do they study the interaction between FXIs and US monetary policy.⁶

We also contribute to the literature on US monetary policy spillovers and the global financial cycle ([Rey 2015](#), [Kalemli-Özcan 2019](#), [Miranda-Agrippino and Rey 2020](#), [Boehm and Kroner 2025](#)).⁷ There is vast empirical evidence of the spillover of US monetary policy to country-level equity markets ([Zhang 2022](#), [Wiriadinata 2021](#), [Dedola et al. 2017](#)), exchange rates ([Gürkaynak et al. 2021](#), [Roussanov and Wang 2023](#), [Eichenbaum and Evans 1995](#), [Faust et al. 2007](#), [Anderson et al. 2003](#)), and interest rates ([Timmer 2018](#), [Zhang 2022](#)). To identify the US monetary transmission channels, [Wiriadinata \(2021\)](#) uses foreign currency debt data, while [Zhang \(2022\)](#) uses currency invoicing shares. However, as discussed in Section 1, using cross-country heterogeneity faces several identification issues due to the presence of country-specific confounding factors and the masking of within-country heterogeneity.⁸

⁵Recent theoretical advances show that interventions can affect the exchange rate and enhance welfare. [Gabaix and Maggiori \(2015\)](#), [Cavallino \(2019\)](#), [Amador et al. \(2020\)](#), [Fanelli and Straub \(2021\)](#), [Maggiori \(2022\)](#), [Itskhoki and Mukhin \(2023\)](#), and [Yago \(2024\)](#) study FXIs under partial segmentation of home and foreign currency bond markets. Since international financial intermediaries have limited risk-bearing capacity, FXIs affect the exchange rate by changing their balance sheet composition. [Hassan et al. \(2023\)](#) show that policies that appreciate domestic currency when the marginal utility of world investors is high increase the market value of firms and stabilize the country's wealth. Hence, small countries optimally choose to stabilize their currencies relative to the US dollar.

⁶[Roussanov and Wang \(2023\)](#) show that global FX dealers buy US dollars in response to contractionary monetary policy shocks, potentially representing the counterpart of the foreign central banks that sell the US dollar in response to contractionary monetary policy shocks.

⁷[Akinci and Queralto \(2024\)](#), [Aoki et al. \(2020\)](#), and [Gopinath and Stein \(2021\)](#) study US monetary policy theoretically and consider corporate or bank balance sheet risk when firms issue debt denominated in dollars.

⁸An exception is [Morais et al. \(2019\)](#), who match loan-level bank lending data with firm-level balance sheet data and study the international risk-taking channel of US monetary policy. While they use monthly loan data, we use daily data on FXIs and firm-level stock prices, as discussed below.

In contrast, we study the spillovers of US monetary policy on stock prices using a cross-section of firms' stock returns. Using daily stock-price data combined with high-frequency US monetary policy shocks allows us to understand the channel through which US monetary policy spills over to other countries. Surprisingly, at least to the best of our knowledge, we are the first to study the cross-section of stock price reactions to US monetary policy in an international context.⁹ The advantage of using firm-level heterogeneity in stock price responses is that it allows us to identify the channels of US monetary policy more cleanly at a high frequency and only exploit differences across firms, controlling for time-variant country-specific observed and unobserved heterogeneity.¹⁰

The majority of literature on international US monetary spillover aggregates high-frequency US monetary shocks to lower frequency and studies their implication on the exchange rate, capital flows, and/or real outcomes (Miranda-Agrippino and Rey 2020). In contrast, we combine daily data on exchange rates, firm stock prices, and FXIs with US monetary surprises. There are several advantages to using daily data as dependent variables. FOMC meetings occur at irregular intervals within each year, and aggregating monetary surprises over each month or quarter can induce serial correlation in aggregate shocks and inconsistent estimates of aggregate impulse responses (Ramey 2016, Anderson and Cesa-Bianchi 2024). Moreover, the magnitude of high-frequency shocks is small. This complicates estimating impulse responses since exchange rates in the distant future are influenced by many other confounding factors. To minimize this “power problem,” we study the response of daily exchange rates and stock prices, which move contemporaneously with monetary shocks (Nakamura and Steinsson 2018).

We also relate to the literature that studies how central banks respond when the US monetary policy tightens. In theory, the central bank should let the exchange rate depreciate (Friedman 1953, Mundell 1957, Fleming 1962). However, in practice, many central banks are reluctant to let the exchange rate fluctuate due to a fear of floating (Calvo and Reinhart 2002), as the evidence for the benefits of a depreciated currency for boosting exports remains elusive (Gopinath et al. 2020). Our results show that the monetary policy-induced depreciation does not benefit exporters as the positive expenditure switching effect is offset by negative demand effects, while at the same time harming firms that have US dollar borrowing.

The IMF integrated policy framework (Basu et al. 2020) studies the interaction between monetary policy and FXIs from a theoretical perspective. They show that after an adverse shock to the foreign appetite for domestic currency debt, FXIs reduce the need for the policy rate to be increased and, in that sense, can enhance monetary autonomy. Yago (2024) shows that by intervening against the Fed, central banks can offset the inflationary pressure driven by a US

⁹A vast literature studies US firm-level equity returns in response to US monetary policy but abstracts away from foreign firms (Gorodnichenko and Weber 2016, Ai et al. 2022, Ozdagli and Velikov 2020, Chava and Hsu 2020, Gürkaynak et al. 2022).

¹⁰The caveat of using stock price data is that we are only able to study public firms, and we cannot see whether a reduction in stock prices eventually materializes in changes in real outcomes, such as employment, domestic revenue, and profitability of those firms, see, e.g., Rodnyansky (2019).

monetary policy-induced currency depreciation. This provides a rationale for why intervening against the Fed can be optimal.¹¹

Finally, our paper is also related to the literature studying financial channels of US monetary policy. In particular, US monetary policy shocks can alter investors' risk perception (e.g., [Bernanke and Kuttner \(2005\)](#), [Bruno and Shin \(2015a;b\)](#)) and global stock market volatility (e.g., [Miranda-Agrippino and Rey \(2020\)](#), [Bauer et al. \(2023\)](#)), the yield curve (e.g., [Nagel and Xu \(2024\)](#), [Hanson and Stein \(2015\)](#), [Timmer \(2018\)](#)), shift portfolio flows (e.g., [Converse and Mallucci \(2023\)](#), [Converse et al. \(2023\)](#)), and widen the currency risk premium (e.g., [Lustig et al. \(2011; 2014\)](#), [Kalemli-Özcan \(2019\)](#)).

3 Data

3.1 Sources

We combine data from several sources. Our sample period is between 2000 and 2019, during which the data on US monetary shocks and corporate balance sheets are available.¹²

First, we collect data on sterilized FXIs based on [Fratzscher et al. \(2019\)](#) and [Adler et al. \(2025\)](#). We use publicly available databases on central bank websites and the FRED database.¹³ Some countries do not publicly disclose the data due to the secret interventions, in which case, we individually contacted the central banks to be granted access to the data. We then restricted the sample countries based on the following criteria: first, to use as much high-frequency data as possible, we only use daily intervention data and exclude countries and periods where only monthly or quarterly data is available. Second, since we study interventions against US monetary shocks, our sample only comprises countries that intervened against the US dollar multiple times during the sample period.¹⁴ The following 13 countries have available data and satisfy the above criteria: Argentina, Australia, Brazil, Chile, Colombia, Costa Rica, Georgia, Hong Kong, Japan, Mexico, Peru, Switzerland, and Turkey.¹⁵ We make the FXI data publicly available for future researchers to build on this paper's findings. One limitation is that the

¹¹[Kalemli-Özcan \(2019\)](#) shows that monetary policy divergence vis-à-vis the United States has larger spillover effects in emerging markets than in advanced economies. Domestic monetary policy is ineffective in mitigating this effect, as the pass-through of policy rate changes into short-term interest rates is imperfect.

¹²Appendix A shows that the result is robust even after excluding special periods, including the global financial crisis and zero lower bound on the interest rate.

¹³We focus on central banks' direct purchases and sales of the US dollar. One may think the signaling channel of FXIs is driving the results, since dollar sales may signal the central bank's concern about the weakness of the domestic currency, which is correlated with future monetary tightening. To address this possibility, Appendix A controls for government bond yield, which reflects future monetary policy expectations, and confirms our results on stock return and exchange rate remain robust. This confirms that the monetary policy expectation channel is not responsible for our results.

¹⁴We focus on US monetary shocks since a limited number of countries disclose daily FXI data against the euro and Japanese yen.

¹⁵To address the possibility that the result is driven by a particular country, in Appendix A, we exclude each country from the regression and show the results on stock prices and exchange rates are robust.

intraday data on FXIs is difficult to obtain. However, even if we use daily data, our result shows that FXIs have persistent effects on the exchange rate and stock market.¹⁶

To address the endogeneity concern of monetary policy, we use the high-frequency change in the Fed funds rate (FFR) identified by [Nakamura and Steinsson \(2018\)](#), which was subsequently updated by [Acosta \(2023\)](#). They estimate the changes in Fed funds futures in a 30-minute window around the FOMC announcement. We obtain daily data on the spot exchange rate and stock price from Thomson Reuters Datastream. Since Datastream reports the closing exchange rate at 4 p.m. London time and the FOMC announcements are made around 2 p.m. Washington time (7 p.m. London time), we moved the reported dates of exchange rates by one date forward so that the exchange rate on each date is reported after the FOMC announcement on the same date. Since the end-of-date stock price is released at different times in different time zones, we adjusted the reported dates for the stock price depending on whether the stock price is released before or after the FOMC announcement.

We use corporate balance sheet data on the Capital IQ platform provided by S&P Global Market Intelligence. The advantage of Capital IQ is that it provides information on the currency denomination of debt, which is not available in other databases such as Worldscope, Compustat, and Orbis. Its Capital Structure database provides detailed information on each debt instrument held by each firm, including the principal amount due, repayment currency, and maturity. For example, Agrometal S.A.I., a manufacturing firm in Argentina, had a total outstanding debt of 5.6 million dollars on December 31, 2015. Among them, 2.2 million dollars are repaid in US dollars, and the remaining 3.4 million dollars are repaid in Argentine pesos. Hence, the share of dollar bonds over total bonds is 39 percent. Capital IQ has provided annual data on corporate balance sheets since 2001. The sample is restricted to publicly listed firms, as the data on stock prices is available.

To measure investors' risk aversion, following the literature ([Bekaert et al. 2013](#), [Bauer et al. 2023](#)), we use the VIX index, which is calculated by the Chicago Board Options Exchange based on the implied volatility in S&P 500 stock index. We construct the currency risk premium using daily exchange rates and interbank rate data available in Bloomberg. The formal definition of the currency risk premium is provided in Appendix B. In addition, we collect the 10-year government bond yield data from Global Financial Data to measure the risk-free rate used for discounting future cash flows. Finally, to study the portfolio rebalancing channel, we use the EPFR equity fund flows dataset, which is available as a country-time panel at a weekly frequency. Investor flows are measured as end investors' purchases and redemptions of shares in mutual funds and ETFs from EPFR Global over the period since 1997, following the approach in [Jotikasthira et al. \(2012\)](#), [Converse and Mallucci \(2023\)](#), [Converse et al. \(2023\)](#). The EPFR flow dataset is a country-time panel at a weekly frequency.¹⁷

¹⁶[Kuersteiner et al. \(2018\)](#) study intraday intervention data in Colombia between 2001 and 2012. However, intraday data is unavailable for the other 12 countries in our sample. See also [Dominguez \(2003\)](#) and [Dominguez et al. \(2013\)](#) for analysis using intraday intervention data.

¹⁷We thank Nathan Converse for his help with the data construction. EPFR provides a comprehensive dataset

We complement the data using a variety of sources. Firm-level data on exports, international sales, and international assets, as well as the incorporation date, are available on Worldscope. To measure firms' reliance on intermediate imports in their production, we use sector-level data on import content of exports in OECD input-output tables, following [Rodnyansky \(2019\)](#). The limitation is that it is difficult to obtain firm-level data on the invoicing currency of exports and imports for many countries. Hence, this paper focuses on the effect of FXIs on firms with dollar debt. Appendix C provides details on the data cleaning procedure and selection criteria for our sample of firms.

Finally, for country-level characteristics, we collect the data on monthly policy rates from the IMF Monetary and Financial Statistics. The data on GDP, inflation rate, trade balance, and unemployment rate are retrieved from the World Bank database and the IMF World Economic Outlook.

3.2 Summary Statistics

This section describes the summary statistics for FFR shocks and FXIs, and the number of sample firms across countries. The detailed tables are available in the appendix.

Table A1 provides summary statistics for FFR shocks and changes in the exchange rate and stock price. Our sample consists of 90 FOMC announcement dates between 2000 and 2019.¹⁸ Row (1) shows the summary statistics for the FFR shock estimated by [Nakamura and Steinsson \(2018\)](#). The FFR shock is defined as the change in market expectations of the Fed funds rate over the remainder of the month in which FOMC meetings occur. The shock is in terms of basis points, and a positive value implies a tightening surprise by the Federal Reserve. As [Nakamura and Steinsson \(2018\)](#) discuss, the magnitude of FFR shocks estimated by the high-frequency method is small: the standard deviation is 1.81 basis points. This “power problem” makes it difficult to estimate the effect on real economic outcomes, such as firm-level investment and economic growth, as the data is available only in quarterly or annual frequency and is affected by many unobservable confounding factors. Hence, we focus on the response of daily exchange rates and stock prices, which move contemporaneously with high-frequency monetary shocks.

Row (2) shows the summary statistics for the exchange rate depreciation, comparing before and after the FOMC announcement date. $e_{c,t}$ is the spot exchange rate at the end of date t in the

on fund-level investor flows into ETFs and mutual funds. Using this data, the flows to individual countries are estimated based on the methodology described in [Jotikasthira et al. \(2012\)](#). Flows into fund f during each week are allocated to country c using fund f 's reported portfolio allocation in country c at the end of the previous calendar year. The data is not available at a daily frequency. Each observation is reported on Wednesday and corresponds to the flows over the seven days until that Wednesday. For example, the April 2 observation reports the flow from March 27 through April 2.

¹⁸[Nakamura and Steinsson \(2018\)](#) and [Acosta \(2023\)](#) report 151 FOMC sample dates between 2000 and 2019. We exclude 46 dates that are characterized by a zero change in the Fed funds rate. This is because our main focus is the unexpected change in the Fed funds rate and the central banks' counteracting intervention against FFR shocks. In Appendix A, we also include those dates in the sample and show that the result is robust. We also excluded 15 FOMC event dates with large FFR shocks so that our results are not affected by outliers (see footnote 20).

country c . The exchange rate is defined as the value of the US dollar in terms of local currency, so that a higher $e_{c,t}$ implies the appreciation of the US dollar or depreciation of local currency. We take the change in the logarithm of the exchange rate from date $t - 1$ to $t + 1$. Similarly, row (3) shows the summary statistics for the percentage change in the stock price, comparing before and after the FOMC announcement. $p_{i,t}$ is the stock price of firm i at the end of date t . The stock price is denominated in local currency.¹⁹ The standard deviations of exchange rate change and stock return are 0.72 percent and 3.47 percent, respectively.²⁰

Table A2 shows the frequency, amount, and sample period of interventions around the FOMC event dates in our sample countries. To consider the possibility that the effects of FFR shocks and FXIs accumulate over time, we consider a 5-day window after FOMC announcement dates.²¹

We first define buying and selling interventions so that central banks buy or sell the US dollar at least once between dates t and $t + 5$, where t is the FOMC announcement date. Columns (1) and (2) report the frequency of buying and selling US dollar interventions. There is a large variation in the frequency of interventions across countries. For example, Argentina intervened 59 times by buying the US dollar and 45 times by selling it, out of a total of 90 FOMC event dates in our sample. In contrast, Switzerland never intervened, and Turkey intervened only once around the FOMC meetings. However, the interventions happened only outside the 5-day window around the meetings. To minimize the possibility that the interventions are affected by a myriad of other factors besides US monetary shocks, our sample does not count interventions that happened outside the 5-day window around FOMC meetings.²²

Column (3) reports the frequency of counteracting interventions, which is the main focus of our analysis. We define counteracting interventions as follows: if the Fed funds rate increases on date t , central banks sell the US dollar at least once and never buy the US dollar between dates t and $t + 5$, and vice versa when the Fed funds rate decreases. Since a higher Fed funds rate depreciates the local currency, central banks offset the depreciation by selling the US dollar and buying the local currency. Unless otherwise stated, we will use this definition of counteracting interventions throughout the regression analyses in this paper. However, our result is robust even if we adopt different definitions of counter-intervention or if we include all intervention dates in our sample.²³ Columns (1)-(3) show that not all interventions are counteracting interventions

¹⁹Appendix A shows that, if the stock price is denominated in foreign currency, the stock price response to Fed tightening becomes larger since changes in exchange rates affect the valuation of stock prices. This implies that firms with dollar debt are riskier investment opportunities for foreign investors than those with local currency debt.

²⁰We trimmed the top and bottom 5% of the FFR shocks and winsorized the top and bottom 5% of changes in the exchange rate and firms' stock prices in each country so that our results are not affected by outliers.

²¹For identification purposes, we focus on a 5-day window around FOMC announcements, since the estimates can be more affected by unobserved confounding factors in a longer time horizon. However, policymakers may be more interested in a longer-lasting effect, as the balance sheet and expenditure switching effects occur at a lower frequency. In Appendix D, we study this lower-frequency effect and find that the effects of US monetary shocks and FXIs on stock prices are persistent for six to eight weeks (1.5 to 2 months).

²²To address the possibility that our result is affected by one particular country with either frequent or infrequent intervention, Appendix A excludes each country from the sample and shows that our result is robust.

²³Appendix A defines the counteracting interventions differently: the central banks' average net sales of US

against the Fed. For example, Argentina intervened by buying the US dollar 59 times and by selling the US dollar 45 times, but only on 15 occasions did they intervene by counteracting the FFR shock. Focusing on these counter-interventions, we will study how the effects of FOMC announcements on the exchange rate and stock price are different when the central banks do and do not counter-intervene against FFR shocks. Columns (4) and (5) report the FXI volume in terms of millions of US dollars around the FOMC announcement dates. The mean and median FXI volumes across all countries are 57 million dollars and 15 million dollars, respectively.²⁴

Column (6) reports the sample period when the FXI data are available. The availability of the FXI data depends on the country. In particular, our sample only includes countries and periods in which daily FXI data is available. We excluded periods in which only monthly, quarterly, or annual data are available. For example, daily intervention data in Switzerland is available only until 2001.

To study how long foreign exchange interventions last, we follow [Fratzscher \(2008\)](#) and [Fratzscher et al. \(2019\)](#) and define the two intervention days as being in the same episode if they are in the same direction and within a ten-trading-day (two-week) time window. Based on this definition, the mean and median lengths of intervention episodes are 10.7 and 7 days, respectively.

Table A3 describes the sample of firms. The table shows the total number of firms and the number of firms that issued dollar debt at least once during the sample period in each country. Our sample consists of 4,060 firms in total, out of which 261 firms (6%) have dollar debt. The average share of dollar debt over total debt across all firms is 66%, conditional on firms issuing a positive amount of dollar debt. The share of firms with dollar debt is relatively large in emerging economies, while most of the Japanese firms do not issue dollar debt as they borrow in Japanese yen (if we exclude Japanese firms from the sample, 14% of firms have dollar debt). As discussed in Appendix A, almost all of the sample firms with dollar debt do not have export revenues and therefore are not naturally hedged.

3.3 Identification of Unexpected FXI

To elaborate on our identification strategy for FXIs, we estimate a central bank reaction function. The motivation is to extract the unexpected component of interventions that cannot be forecasted by FFR shocks, past exchange rate movements, FXIs before the FOMC events, and other macroeconomic characteristics. This is a popular approach to minimizing the endogeneity of interventions in the literature ([Kearns and Rigobon 2005](#), [Ito and Yabu 2007](#), [Fatum and Hutchison 2010](#), [Kuersteiner et al. 2018](#), [Fratzscher et al. 2019](#)), similar to residuals from a

dollars between dates t and $t + 5$ is positive when the Fed funds rate increases at date t , and vice versa when the Fed funds rate decreases. Our main results are robust to using this alternative definition.

²⁴To calculate the mean and median FXI volumes around FOMC announcement dates, we first take the average of absolute values of FXI volumes between date t and $t + 5$ for each FOMC announcement date t . Next, we take the mean and median of implied FXI volume over FOMC announcement dates, conditioning that there was at least one intervention between date t and $t + 5$.

monetary policy Taylor rule. The advantage of the deviation from the FX policy rule relative to deviations from a monetary policy rule is that FXIs vary on a daily level, while monetary policy decisions are usually only conducted every several weeks. Therefore, the high-frequency FXI rule approach more cleanly identifies surprises than the Taylor rule residual.²⁵

All of our results are robust across various definitions or measures of FXIs. If we do not use indicators but instead adopt continuous measures for the raw volume (size) of FXIs, or if we do not take the surprise component by estimating the FXI rule and include all intervention dates in our sample, our key economic implications on exchange rates and stock prices do not change. However, we use the following definition of FXIs as a benchmark to better identify the direct effects of FXIs that are less likely to be confounded by other factors.

To do so, we consider the following FXI rule:

$$\widetilde{FXI}_{c,t} = \sum_c \beta_c (FFR_t \times \gamma_c) + \delta Z_{c,t} + \gamma_c + \epsilon_{c,t}. \quad (1)$$

$\widetilde{FXI}_{c,t}$ is the indicator for counteracting intervention in country c on FOMC announcement date t , as discussed in column (3) of Table A2, as is standard in the literature.²⁶ $\widetilde{FXI}_{c,t}$ takes 1 if the Fed tightens unexpectedly on date t and the central banks intervene by selling the US dollar at least once, but never intervene by buying the US dollar between dates t and $t + 5$. Similarly, $\widetilde{FXI}_{c,t}$ takes -1 if the Fed loosens unexpectedly on date t and the central banks intervene by buying the US dollar at least once but never intervene by selling the US dollar between dates t and $t + 5$. Otherwise, $\widetilde{FXI}_{c,t}$ takes zero. In Section 5.1,

FFR_t is the FFR shock in terms of basis points, and γ_c is the fixed effect for each country c . Their interaction $FFR_t \times \gamma_c$ captures the differential propensity to intervene against FFR shocks across countries. $Z_{c,t}$ is the set of controls, including the trend and standard deviation of the exchange rate and the dummy for FXIs before the FOMC event date, as well as the macroeconomic variables (one-month lagged policy rate, one-year lagged GDP, CPI inflation rate, trade balance over GDP ratio, and unemployment rate), and the interaction of macroeconomic variables with FFR shock.²⁷ For past exchange rate movement, we took the percentage change and standard deviation of the exchange rate between dates $t - 1$ and $t - 5$, where t is the FOMC event date. For past interventions, the dummy takes 1 if the average net purchase of US dollars

²⁵This approach is conceptually similar to the estimation of the Taylor rule residual for monetary policy (Taylor 2009, Maddaloni and Peydró 2011, Dell’Ariccia et al. 2017). In a similar vein, Christiano et al. (1996; 1999) estimate a structural vector autoregression (VAR) model to identify monetary policy shocks, and Blanchard and Perotti (2002) estimate a VAR model to identify fiscal policy shocks.

²⁶We use an indicator variable for FXIs following the literature on FXI reaction function: Fatum and Hutchison (2010) and Fratzscher et al. (2019) (Appendix II) define a (0,1) indicator for intervention and no intervention, and Ito and Yabu (2007) define a (0,1,-1) indicator for buying and selling dollar interventions. Ito and Yabu (2007) use an FXI indicator instead of volume because the FXI volume is determined within the day depending on intraday exchange rate movement, but intraday interventions are not disclosed. By using the indicator variable, they mitigate this endogeneity concern caused by our inability to estimate the intraday reaction function.

²⁷We use a monthly policy rate since the daily policy rate is not available for all sample countries. In Appendix A, we use daily policy rates in countries with available data and show that our result is robust.

between dates $t - 1$ and $t - 5$ is positive, -1 if the net purchase is negative, and zero if there are no interventions.

We follow the previous empirical literature on FXIs regarding the estimation of an FXI policy rule. Following [Fratzscher et al. \(2019\)](#), we control for the past exchange rate trend and volatility and past interventions, as they can affect the central bank's decision to intervene. Moreover, based on [Fatum and Hutchison \(2010\)](#), we also control for macroeconomic variables, such as GDP and trade balance, since countries with different macroeconomic conditions may adopt different interventions. We use lagged macroeconomic variables to remove the simultaneity bias. We also account for the interaction between the FFR shock and macro variables, since countries with different macroeconomic characteristics can respond heterogeneously to the FFR shock. We include the country fixed effects, γ_c , to control for the difference in average exchange rate trends in each country.

The predicted counter-intervention from estimating the reaction function (1) can be interpreted as the expected component of counter-interventions, in other words, the average response of FXIs to FFR shocks, past exchange rates, interventions, and macroeconomic conditions. The residual, or the deviation from the FXI rule, can be interpreted as the unexpected component of FXIs. We exploit this residual as the exogenous surprise component of FXIs.

The appendix provides a graphical illustration of the results from estimating the FXI policy rule. Figure A1 shows the result for a variance decomposition from estimating Equation (1). Our estimates show that 24% of the variation in counter-intervention can be explained by the set of controls ($R^2 = 0.24$), while the remaining 76% is unexplained. This low R-squared implies that a large part of FXIs cannot be predicted by the FFR shock, past exchange rates, or interventions. If we further decompose the controls, 16% can be explained by the FFR shocks, 2% by macroeconomic variables (policy rate, GDP, CPI inflation rate, trade balance over GDP ratio, unemployment rate), 30% by the interaction between the FFR shock and macro variables, 35% by past interventions, and 17% by country fixed effects. The contribution of past exchange rate trends and volatility is almost zero, so it is not displayed in the figure. This is consistent with [Fratzscher et al. \(2019\)](#), who show that past exchange rates have very limited explanatory power for FXIs.

Having shown that the variation of interventions is difficult to predict, we will use the residual from estimating Equation (1) as an unexpected component of counter-interventions. To simplify the interpretation of results in later sections, we define an unexpected counter-intervention dummy, $FXI_{c,t}$. If the residual from estimating Equation (1) is greater than its median in absolute value, $FXI_{c,t}$ takes one, and central banks counter-intervene unexpectedly against the Fed. Otherwise, $FXI_{c,t}$ takes zero, and central banks do not counter-intervene unexpectedly. Although we use this indicator as the benchmark, we emphasize again that our results are robust even if we do not take dummies but instead the continuous measure for the raw volume of FXIs.

Figure A2 shows a simple graphical example of this identification methodology. Panel

(a) shows an example of unexpected US monetary tightening: on November 15, 2000, the Fed tightened unexpectedly, and the Reserve Bank of Australia intervened by selling the US dollar. The counter-intervention dummy $\widetilde{FXI}_{c,t}$ takes 1. The predicted value and residual from estimating the linear probability model (1) are around 0.2 and 0.8, respectively. This implies that the market expected that there was a 20% probability that the central bank would intervene against the Fed's tightening shock by selling the US dollar, but an 80% probability that the central bank would not intervene. This large residual implies that the interventions were mostly unexpected. Similarly, panel (b) shows the example of unexpected US monetary easing: on March 22, 2005, the Fed delivered an accommodating monetary policy shock, and the Central Bank of Argentina intervened by buying the US dollar. The counter-intervention dummy $\widetilde{FXI}_{c,t}$ takes -1 . The predicted value and residual from estimating the linear probability model (1) are around -0.39 and -0.61 , respectively. This implies that the market expected that the probability that the central bank would not intervene against the US monetary easing shock was more than 60% (the residual is 0.61 in absolute value). This suggests that the degree to which the interventions are unexpected can be measured using the absolute value of the residual from the estimating equation (1). In the baseline analysis, we define the interventions as unexpected if the residual is larger than its median in absolute value. Our result is robust even if we use different criteria for defining unexpected counter-intervention or use the continuous size of FXIs and include all intervention dates without taking the unexpected component.

The advantage of our identification approach is that, by estimating the residual from an FXI policy rule, we can extract the surprise component of FXI and study its effect on the exchange rate. Our approach, however, is not without limitations. It is difficult to find a perfect instrument since unobserved confounders, such as investors' sentiments or expectations, can affect FXIs, exchange rates, and stock returns at the same time.

Future literature is encouraged to explore other identification strategies. One possible avenue for research is the narrative approach. Literature on monetary policy, which was pioneered by [Friedman and Schwartz \(1963\)](#) and later developed by [Romer and Romer \(1989\)](#), uses historical FOMC records or news articles to identify monetary policy decisions that were not driven by inflation or output.²⁸ The narrative approach is powerful but challenging to apply to FXI, since intervention decisions are often made in secret and not publicly disclosed.²⁹ Future availability of FXI data and firm-level currency exposure would improve the accuracy of our estimates and enable more comprehensive analyses of the effect of FXI on the exchange rate and the stock market.

²⁸See [Romer and Romer \(2023\)](#) for developments of the narrative approach and its application to recent data.

²⁹An exception is [Naef \(2024\)](#), who hand-collects confidential daily reports on intervention decisions by the Bank of England from 1987 to 1992 to identify the surprise component of FXI.

4 Empirical Strategy

Our empirical strategy relies on a high-frequency event-study approach that examines the performance of equities and exchange rates around FOMC meetings. The event study approach has the advantage that the market reaction on FOMC dates is likely due to monetary policy itself, rather than other confounding factors that could influence equity prices or the exchange rate. For instance, in a simple time-series regression in which quarterly outcome variables are regressed on US monetary policy (shocks), it is more difficult to identify the causal effect of monetary policy as many confounding factors could be the reason for the market reaction that is not due to monetary policy itself. If the monetary policy shock is completely exogenous, the coefficient may not be biased, but aggregating high-frequency monetary policy shocks to the quarterly level may cause a “power issue”, similar to a weak instrumental variable problem in two-stage least squares regressions.

We start with event-study local projections (Jordà 2005) by estimating the following sequence of regressions across FOMC dates:

$$y_{i(c),t+h} - y_{i(c),t-1} = \beta_h FFR_t + \mathbf{X}\delta_x^h + \alpha_{i(c)}^h + \epsilon_{i(c),t}^h, \quad \forall h \in [-5, 5] \quad (2)$$

where FFR_t is the FFR shocks developed by Nakamura and Steinsson (2018) and Acosta (2023). The shock is defined as the change in the Fed funds futures rate in a 30-minute window around the FOMC announcement. FFR_t is in terms of basis points, and a positive FFR_t represents the unexpected increase in the Fed funds rate. $y_{i(c),t+h}$ is the logarithm of stock price of a firm i based in the country c , h days after the FOMC meeting. $\alpha_{i(c)}^h$ is a firm fixed effect. The standard errors are always double-clustered at the firm and event date level to account for correlation at the same firm and time. β_h is the effect of the FFR shock on the equal-weighted stock price h days after the FOMC meeting. Equation 2 is informative about the spillover effects of US monetary policy across all countries in our sample and across firms. $\beta < 0$ for each $h \geq 0$ implies that a surprise tightening of US monetary policy reduces stock prices abroad h days after the meeting. Then we can estimate Equation 2 for the country-FOMC date subsamples with and without unexpected counter-intervention ($FXI_{c,t} = 1$ and $FXI_{c,t} = 0$). For the subsample $FXI_{c,t} = 1$, $\beta = 0$ implies that monetary policy does not spill over negatively to countries’ equally-weighted stock price index h days after the meeting if they counter-intervene against the Fed. $\mathbf{X}$ is a set of controls that include the one-year lagged export intensity, total asset, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interactions with FFR shock. The interaction of the Fed funds rate with the firm-level characteristics ensures that the differential responsiveness across firms to changes in interest rates is not responsible for the identified effect of the balance sheet channel. We consider two modifications to evaluate the effect of FXIs. First, we introduce an interaction term between the

Federal funds rate shock and dollar debt:

$$y_{i(c),t+h} - y_{i(c),t-1} = \gamma_h \text{FFR}_t \times \text{USD}_{i(c),y-1(t)} + \mathbf{X} \delta_x^h + \alpha_{i(c)}^h + \alpha_{c,t}^h + \epsilon_{i(c),t}^h, \quad \forall h \in [-5, 5] \quad (3)$$

γ_h can be interpreted as a state-dependent effect of the monetary policy shock, where the state is whether firms have US dollar debt or not (Cloyne et al. 2023). $\text{USD}_{i(c),y-1(t)}$ is a dummy if a firm has had US dollar debt in the previous year before the FOMC meeting. Appendix A shows that our results are robust even if we do not take a dummy but instead use a continuous measure for dollar debt. $\alpha_{c,t}^h$ is a country-time fixed effect and $\alpha_{i(c)}^h$ is a firm fixed effect. γ captures the balance sheet channel of US monetary spillover. $\gamma < 0$ for each $h \geq 0$ implies that a surprise tightening of US monetary policy reduces the stock price of firms with dollar debt relative to those without dollar debt. Note that the inclusion of the country-time fixed effects controls for all time-varying observed and unobserved characteristics, and only exploits the effects across firms within a country-time dimension. The inclusion of the fixed effects also implies that all other variables that are spanned by the fixed effects are collinear and cannot be estimated, e.g., FFR_t itself.

Next, we introduce an interaction term between the Federal funds rate shock and the intervention:

$$y_{i(c),t+h} - y_{i(c),t-1} = \beta_h \text{FFR}_t + \Omega_h \text{FXI}_{c,t} + \gamma_h \text{FFR}_t \times \text{FXI}_{c,t} + \mathbf{X} \delta_x^h + \alpha_{i(c)}^h + \alpha_{c,t}^h + \epsilon_{i(c),t}^h, \quad \forall h \in [-5, 5] \quad (4)$$

γ captures the effect of FXIs mitigating the effect of US monetary spillover. $\gamma > 0$ for each $h \geq 0$ implies that the decline in stock price is smaller with FXIs than the case without FXIs. In contrast to standard local projections, we also consider $h < 0$ in the spirit of an LP-DID proposed by Dube et al. (2023). One difference between the LP-DID and the standard DID is that a sequence of regressions is estimated for each h . This has the advantage that β_h is unaffected by the choice of the number of lags and leads included. Moreover, the LP-DID avoids several other problems compared to estimating a difference-in-differences specification with two-way fixed effects, see, e.g., Callaway and Sant'Anna (2021), Goodman-Bacon (2021) among many others.

For the difference-in-differences estimator to be unbiased, we require the parallel trend assumption to be satisfied—that is, absent a shock, treated and control firms would have evolved the same way. While it is not possible to test this assumption, as the counterfactual post-FOMC behavior without the shock is unobservable, we can test whether there are differential pre-trends before the shock. Estimating β_h for $h < 0$ allows us to test whether there is a violation of the parallel trend assumption.

Recent literature has argued that DID designs are likely to be biased in the presence of a staggered DiD approach, as already treated units can act as effective comparison units (Baker et al. 2022). Note that this is not a concern in our setting as we set $h \in [-5, 5]$, covering

only a window of 10 days, which prevents overlapping observations and staggered treatment, as FOMC meetings only occur approximately every six weeks. The concern would be that firms with US dollar debt are treated for one FOMC meeting but not for the next, but would still be treated as comparison units for the next one.

We start by estimating Equations 2 and 3 separately for FOMC meetings when country c counter-intervenes in the FX market after the FOMC meeting, based on the definition in Section 3. Focusing on no-counter-intervention events allows us to test the degree and channels of US monetary policy spillovers if countries do not intervene against the Fed. Instead, focusing on counter-intervention events allows us to test whether US monetary policy spills over to firms' stock prices if a country intervenes against the Fed. In particular, if we can reject the null hypothesis that $\beta = 0$ in Equation 2, the data favors the alternative hypothesis that there exists a spillover effect of US monetary policy. If we cannot reject $\beta = 0$, there are likely no spillover effects. In Equation 3, the null hypothesis is that firms with US dollar debt are not differentially affected by a US monetary spillover, while the alternative hypothesis is that they are differentially affected. The alternative hypothesis, $\gamma < 0$, can be interpreted as a US monetary policy-driven balance sheet channel of depreciation.

The disadvantage of splitting the data into intervention and no-intervention events is that we cannot test whether β and γ are statistically different for intervention and no-intervention events. It is possible that when estimating the equations separately, we can reject the null hypothesis in one subsample but not in the other, yet due to large standard errors, the two situations are not different from each other in a statistically significant manner.

We therefore refine our regression equation by including the counter-intervention dummy specifically in the regression equation instead of splitting between situations when $FXI_{c,t}$ is either zero or one:

$$y_{i(c),t+h} - y_{i(c),t-1} = \theta_h FFR_t \times USD_{i(c),y-1(t)} \times FXI_{c,t} + \gamma_h FFR_t \times USD_{i(c),y-1(t)} + \mathbf{X}\delta_x^h + \alpha_{i(c)}^h + \alpha_{c,t}^h + \epsilon_{i(c),t}^h, \quad \forall h \in [-5, 5] \quad (5)$$

In Equation 5, the coefficient γ has the same interpretation as in the equation without the triple interaction (Equation 3) when estimating in the sample of no-intervention: the relative performance of firms with US dollar debt in response to a contractionary US monetary policy shock when the country does not counter-intervene. A negative coefficient implies that firms with US dollar debt underperform relative to those with US dollar debt. The triple interaction coefficient θ measures the marginal effect of FXIs for firms with US dollar debt in response to a contractionary monetary policy shock. A positive coefficient θ implies that FXIs lead to relatively higher stock prices for firms with US dollar debt in response to a contractionary monetary policy shock, compared to a counterfactual under which the central bank does not counter-intervene. θ can therefore be interpreted as the extent to which FXIs mute the US monetary policy-induced balance sheet channel of exchange rate depreciation, while γ measures

the extent of the balance sheet channel without interventions. Summing θ and γ can be equivalently interpreted as Equation 3 for $FXI_{c,t} = 1$ as the balance sheet channel when countries intervene. $\theta + \gamma = 0$ implies FXIs entirely mute the balance sheet channel.

Note that estimating the relative stock market response of having US dollar debt in response to the monetary policy shock allows us to saturate the regression specification with country-time fixed effects ($\alpha_{c,t}$). Country-time fixed effects control for time-variant observed and unobserved characteristics at the country level, such as the movement of the exchange rates or the effect on the average stock price around the US FOMC meetings. The inclusion of country-time fixed effects implies that the effect of FFR_t is not identified, as it is collinear with the fixed effects (note that $\alpha_{c,t}^h$ absorbs the terms FFR_t , $FXI_{c,t}$, and $FFR_t \times FXI_{c,t}$). Hence, when controlling for time-variant observed and unobserved characteristics at the country level through country-time fixed effects, we can only make a relative statement about having US dollar debt. In an alternative specification, we remove the country-time fixed effects from the regression specification to evaluate the total effect of US monetary policy shocks for both firms with and without dollar debt, with the caveat of controlling for fewer potential confounding factors.

5 Results

We now turn to the empirical result. We study different transmission channels through which US monetary policy affects the foreign equity market and how FXIs modify these effects. First, Section 5.1 establishes that “intervening against the Fed” successfully mitigates the US monetary spillover via the balance sheet channel, exploiting firm-level balance-sheet heterogeneity for identification purposes. Next, Sections 5.2 and 5.3 study how US monetary policy and FXIs affect the exchange rate, risk premium, and portfolio flows. Section 5.4 shows that FXIs can harm exporters via the expenditure switching channel and discusses the central banks’ tradeoff between export competitiveness and financial stability. Section 5.5 measures how much volume of FXIs is sufficient to offset the US monetary transmission to the stock market. Section 5.6 shows that FXIs are more effective in countries with large foreign exchange (FX) reserves. Finally, Section 5.7 provides robustness checks.

5.1 Stock Market Transmission via the Balance Sheet Channel

We begin by estimating Equation 2 separately for firms with and without US dollar debt, i.e., meaning that $(USD_{i(c),y-t(t)} = 1 \text{ and } 0)$. Table 1 shows the result, where panels (a) and (b) show the result with and without FXIs, respectively. We first study the case without FXIs ($FXI_{c,t} = 0$). Panel (a), columns (1) and (2) report the estimated β_1 coefficient for a subsample of firms with and without dollar debt, respectively, based in countries without FXIs. Column (1) shows that, if central banks do not counter-intervene against the Fed, an unexpected increase in the Fed funds rate reduces the stock price for firms with dollar debt in a statistically significant

manner (6.6% in response to a 10 bp surprise hike).³⁰³¹ However, if firms do not have dollar debt, the decline in stock price is mitigated (0.9%). Next, to test if the difference in the response between firms with and without dollar debt is significant, we estimate Equation 3 in countries without interventions. Column (3) reports the estimated β_1 and γ_1 coefficients for countries without interventions. The negative γ_1 coefficient shows that when the Fed funds rate increases, the decline in stock prices for firms with dollar debt is significantly larger than for those without dollar debt (3.1pp larger decline). Moreover, in column (4), we include the country-time fixed effect $\alpha_{c,t}^h$, which captures the time-varying observable and unobservable characteristics at the country level, and our result remains robust. Note that $\alpha_{c,t}^h$ absorbs the FFR_t standalone term. These results suggest a strong negative balance sheet channel driven by US monetary tightening without FXIs.

In contrast, panel (b) conducts a similar exercise with FXIs ($FXI_{c,t} = 1$). Column (1) shows that, if central banks counter-intervene against the Fed's surprise hike, the decline of stock prices is mitigated even if firms have dollar debt (2.2% in response to a 10 bp surprise hike, in contrast to 6.6% in panel (a)). Columns (3) and (4) show the differential response of firms with and without dollar debt. The coefficient γ_1 is small and statistically insignificant, implying that if central banks counter-intervene against the Fed, the stock price response of firms with dollar debt is not different from those without dollar debt in a statistically significant manner. This suggests that FXIs can successfully mute the negative spillover of US monetary shocks via the balance sheet channel, disproportionately benefiting firms with dollar debt.

In addition, comparing column (3) of panels (a) and (b), the β coefficient on the FFR shock is negative and bigger in magnitude with FXIs ($\beta = -0.158$) than without FXIs ($\beta = -0.097$). This suggests the possibility that FXIs may harm firms without dollar debt. In Section 5.4, we test whether this negative effect can be explained by firms with international revenues and assets.

Next, to test whether the balance-sheet effect of FXIs is large in a statistically significant manner, we estimate Equation 4 separately for firms with and without dollar debt. Columns (1) and (2) of Table A4 report the estimated β_1 and γ_1 coefficients for firms with and without

³⁰As shown in Table A1, the standard deviation of the monetary shock in our sample is very small (1.81 bp) since it is measured in a 30-minute window around the FOMC announcement. In fact, in the dataset of [Nakamura and Steinsson \(2018\)](#), there are only six FOMC announcement dates in our sample period (between 2000 and 2019) when the magnitude of the FFR shock is greater than or equal to 10 bp in absolute value. Hence, a 10 bp monetary shock can be interpreted as a large shock.

³¹One may think this decline in stock prices is driven by the portfolio rebalancing channel: a surprise US tightening lowers the US equity return and raises its expected returns, which induces US investors to shift funds to US assets. We would expect that this is not the main driver of our results. While the portfolio rebalancing channel can explain the overall decline in local stock returns, it is less likely to explain differential stock returns across firms, unless international investors disproportionately hold certain firms. One may potentially expect US investors to have a higher exposure toward larger firms, and the portfolio rebalancing channels are driven by large firms. To take this size effect into account, we control for firm size and its interaction with the FFR shock. This disentangles the balance sheet and portfolio rebalancing channels. Finally, Appendix A.1 exploits the debt maturity structure for identification. It is unlikely that international investors hold firms with differential maturity structure, even if they disproportionately hold firms with US dollar debt.

dollar debt, respectively. Column (1) shows that, if firms have dollar debt, a surprise hike in the Fed funds rate reduces the stock price without interventions (6.4% in response to a 10 bp hike), implying a negative balance sheet channel of US monetary policy. However, if central banks intervene, the decline in the stock price is mitigated by 4.5bp, implying that FXIs can mitigate this balance sheet channel. In column (2), we conduct a similar exercise for firms without dollar debt and find that the effects of the US monetary shock and FXIs are small. Finally, in columns (3) and (4), we estimate the triple interaction (Equation 5) to study whether the effect of FXIs is greater for firms with dollar debt. The negative coefficient on $FFR_t \times USD_{i(c),y-1(t)}$ ($\gamma = -0.310$) implies a negative balance sheet channel due to US monetary policy spillovers. However, the coefficient on $FXI_{c,t} \times FFR_t \times USD_{i(c),y-1(t)}$ is positive ($\theta = 0.324$) and $\theta + \gamma = 0$ holds statistically. This result implies that FXIs entirely mute the balance sheet channel. The finding is robust even after including the country-time fixed effect (column 4). Note that $\alpha_{c,t}^h$ absorbs the terms FFR_t , $FXI_{c,t}$, and $FFR_t \times FXI_{c,t}$. This key implication is robust even if we do not take a dummy but instead use a continuous measure of FXIs, and even if we do not take a surprise component of FXIs but include all intervention dates in our sample.

To test whether the effect of FXIs is persistent over time, we estimate Equation 2 over a 5-day window around the FOMC announcement. Figure 2, panel (a) plots the estimated coefficient β_h for all $h \in [-5, 5]$. The red and blue lines show the results for countries with and without FXIs, respectively. Without FXIs, a 10 basis point surprise increase in the Fed funds rate leads to an immediate decline in stock prices, and this accumulates up to nearly 10% after three days, suggesting that the balance sheet channel of US monetary shocks is persistent. However, if countries intervene, the effect of the US monetary surprise is smaller, and it disappears five days after the shock. Importantly, the near-zero coefficients for $h < 0$ suggest that there is little difference in the pre-trends of stock prices in countries with and without FXIs, potentially suggesting that the post-FOMC differential response of stock prices is driven causally by FXIs. To test if the effect of FXIs is large in a statistically significant manner, in panel (b), we estimate Equation 4 and plot the coefficient γ_h for all $h \in [-5, 5]$. $\gamma_h > 0$ for $h > 0$ implies that FXIs can successfully mute the persistently negative balance sheet effects of US monetary shocks. In panels (c) and (d), we repeat a similar exercise for firms without dollar debt. The graph shows that the US monetary shock and FXIs have little effect on the stock price of firms without dollar debt. Finally, to compare the effect of FXIs on firms with and without dollar debt, we estimate Equation 5 for all $h \in [-5, 5]$. Figure 3 plots the estimated coefficient θ_h . $\theta_h > 0$ for $h > 0$ suggests that the effect of FXIs is persistently greater for firms with dollar debt.

5.2 Risk Premium and Portfolio Rebalancing

We have discussed how monetary policy affects stock prices through a balance sheet channel. However, the balance sheet channel may not be the only mechanism through which US monetary policy shocks affect foreign stock prices. In this section, we dig into how monetary policy affects

stock prices through financial channels other than the balance sheet, including changes in the currency risk premium as well as portfolio rebalancing. In particular, US monetary policy shocks can alter investors' risk perception (e.g., [Bernanke and Kuttner \(2005\)](#), [Bruno and Shin \(2015a;b\)](#)) and global stock market volatility (e.g., [Miranda-Agrippino and Rey \(2020\)](#), [Bauer et al. \(2023\)](#)), the yield curve (e.g., [Nagel and Xu \(2024\)](#), [Hanson and Stein \(2015\)](#), [Timmer \(2018\)](#)), shift portfolio flows (e.g., [Converse and Mallucci \(2023\)](#), [Converse et al. \(2023\)](#)), and widen the currency risk premium (e.g., [Lustig et al. \(2011; 2014\)](#), [Kalemli-Özcan \(2019\)](#)), while also influencing the US dollar. Each of these channels can, in turn, affect the pricing of foreign equities.

Below, we first test how US monetary policy affects those variables and whether FXIs mitigate them. We then show how the movements of those variables affect the pricing of foreign equity and differentially so across firms with and without US dollar debt.

5.2.1 US Monetary Policy Transmission

In the first step of our analysis, we study how US monetary policy transmits abroad via financial channels other than the balance sheet channel. We re-estimate Equation 2, where now the dependent variable is the VIX, the currency risk premium, foreign yields, and portfolio flows.

To begin with, US monetary policy shocks affect investors' risk appetite, measured by the VIX, and drive the "Global Financial Cycle" ([Rey 2015](#), [Miranda-Agrippino and Rey 2020](#)). A contractionary monetary policy shock increases the investors' risk aversion, which in turn leads to global capital retrenchment and stock price declines. Following the literature ([Bekaert et al. 2013](#), [Bauer et al. 2023](#)), we use the VIX as an indicator of risk aversion.

Furthermore, US monetary tightening not only increases investors' risk aversion in the stock market but also reduces investors' willingness to assume foreign currency risk. Since the dollar tends to appreciate in "bad" times, foreign currency assets with lower returns become less attractive because they do not sufficiently hedge against increased marginal utility. In stochastic discount factor (SDF) terms, the negative covariance between the asset's return (here, the foreign currency return) and the SDF increases, leading investors to demand a higher excess return, i.e., a risk premium on foreign currency exposures. This elevated risk premium, in turn, lowers the stock price today relative to the future.^{32,33}

Figure 4, panel (b) shows that, without intervention, the currency risk premium increases in response to contractionary monetary policy shocks, but the currency risk premium does

³²Classical evidence shows that a US monetary tightening leads to a persistent dollar appreciation and higher excess return on the US dollar ([Eichenbaum and Evans 1995](#)). In contrast, modern evidence shows that an increase in the US interest rate forecasts an immediate appreciation and a subsequent depreciation of the dollar, implying higher excess return on the foreign currency ([Engel 2016](#)). Literature explains this pattern using a model where the US supplies safe assets during crisis times ([Maggiori 2017](#), [Hassan et al. 2023](#), [Jiang et al. 2021; 2024](#)).

³³Note that this relationship only holds for currencies with a positive risk premium. In the following analysis, if we run a regression for the Japanese yen and Swiss franc, which have a negative currency risk premium, the results do not hold.

not increase when countries counter-intervene. In contrast to the VIX, we can leverage cross-country data to see whether the foreign currency risk premium varies across countries. In particular, FXI can mitigate the exchange rate risk and reduce the volatility of the foreign currency return. Hence, investors become more willing to invest in foreign currencies and require less compensation for taking foreign exchange risks. In SDF terms, FXI decreases the negative covariance between the foreign currency return and the SDF, making investors require a lower risk premium on foreign currency exposures.³⁴

In addition, US monetary policy shocks may influence foreign stock prices via changes in the yield curve. Without interventions, a US monetary tightening shock depreciates the foreign exchange rate, raising the expected foreign short-term policy rate. This higher expected short-term interest rate can also raise the entire foreign yield curve—thereby raising the risk-free rate used for discounting foreign stock cash flows (Hanson and Stein 2015, Nagel and Xu 2024).³⁵ FXIs can stabilize the exchange rate and thus mitigate the international spillovers of US monetary policy via changes in foreign yield curves. To proxy for this foreign currency risk-free rate channel, we employ 10-year sovereign yields. Empirically, the foreign bond yield channel is relatively muted and not much affected by FXIs, as can be seen in Figure 4, panel (c).

Lastly, contractionary US monetary policy may prompt withdrawals from foreign currency portfolios, depressing stock prices through a portfolio rebalancing channel. We test this channel using investor flows to the countries in our sample around FOMC announcements. Panel (d) shows strong investor outflows from foreign funds as a response to a contractionary US monetary policy. However, consistent with the idea that FXIs reduce currency risk, those outflows are mitigated when the foreign country intervenes against the Fed.

5.2.2 Stock Market Transmission

These results raise the question of whether US monetary policy not only affects firms' stock prices through a balance sheet channel, by preventing the reduction in the net worth, but also can work through a risk premium channel or a portfolio rebalancing channel. In the next step of the analysis, we test whether foreign stock prices are indeed affected through those channels and how they affect our coefficients. As can be seen from Table 1, panel (a), without interventions, even firms without US dollar debt are affected by the US monetary policy shock. The channels outlined above should indeed affect those firms, while firms with US dollar debt are additionally affected through the balance sheet channel. In the next step of the analysis, we test whether foreign stock prices are indeed affected through the risk premium and portfolio rebalancing

³⁴In Figure 4, we separate the sample into dates with and without interventions. In Figure A3, we instead take the interaction between the FFR shock and intervention and confirm that the effects of FXI on the risk premiums and capital flows are statistically significant.

³⁵Hanson and Stein (2015) explain this channel using a model where investors' objective function depends not only on future expected returns but also on current income or yield. An increase in the short-term rate prompts investors to rebalance their portfolio from long-term to short-term bonds, generating upward pressure on long-term bond yields.

channels and how they affect our coefficients.

First, we restrict the sample to countries without interventions and study stock market spillovers of US monetary policy via risk premium and portfolio rebalancing channels. To do so, we include the VIX, currency risk premium, foreign bond yield, and equity portfolio flows as controls δ_d^h in Equation 2:

$$y_{i(c),t+1} - y_{i(c),t-1} = \beta \text{FFR}_t + \mathbf{X}\delta_x + \mathbf{D}\delta_d + \alpha_{i(c)} + \epsilon_{i(c),t} \quad (6)$$

Hence, we estimate equation Equation 6 for firms with and without US dollar debt separately and in countries without interventions. As shown in Table 2, panel (a) (column 1), for firms without US dollar debt, the contractionary US monetary policy shock affects those firms' stock prices negatively as long as we do not control for the other channels. However, once we control for the above channels, the coefficient on the US monetary policy shock becomes insignificant (columns 2-6). In particular, currency risk premium and the equity risk aversion channel explain the negative effect of the US monetary policy shocks on firms' stock prices, while the effects of bond yield and portfolio rebalancing channels are muted. Instead, panel (b) of the same table focuses on the firms with US dollar debt. For those firms, the coefficient on the FFR shock decreases slightly once we control for the other channels, but remains statistically significant, suggesting the balance sheet channel plays an important role for those types of firms.³⁶

Next, we test whether those channels interact with the effectiveness of FXIs. We study whether the effect of FXIs on firms with dollar debt can solely be attributed to the balance sheet channel or also to the risk premium and portfolio rebalancing channels. To do so, we restrict the sample to firms with dollar debt and re-estimate Equation 4, controlling for other channels:

$$y_{i(c),t+1} - y_{i(c),t-1} = \beta \text{FFR}_t + \Omega \text{FXI}_{c,t} + \gamma \text{FFR}_t \times \text{FXI}_{c,t} + \mathbf{X}\delta_x + \mathbf{D}\delta_d + \alpha_{i(c)} + \epsilon_{i(c),t} \quad (7)$$

As shown in Table 3, column (1), if we do not control for other channels, the FFR shock lowers the stock price of firms without dollar debt, and FXIs mitigate this stock price decline. Columns (2)-(6) show that, when we control for other channels, both the coefficients on FFR_t and $\text{FFR}_t \times \text{FXI}_{c,t}$ become slightly smaller in magnitude. However, the estimates remain statistically significant, and we cannot reject the null hypothesis that the estimates in column (1) and columns (2)-(6) are about the same.

In sum, “intervening against the Fed” mitigates stock price declines not only via the balance sheet channel but also via the currency risk premium and portfolio rebalancing channels. However, even after controlling for the currency risk premium and portfolio rebalancing channels, the balance sheet channel of FXI remains intact.

³⁶This can also be seen in Table A5, which shows the results for the whole sample and focuses on the interaction effect between the FFR shock and dollar debt. The baseline coefficient diminishes once we include those controls, while the coefficient on the interaction term remains statistically significant. This indicates that the “direct” effect of contractionary US monetary policy shock goes through the above channels, while the balance sheet channel remains statistically significant, explaining the differential effect.

5.3 Exchange Rate

To further understand the balance sheet channel, we study the effect of FXIs on the exchange rate. Our previous results suggest that US monetary tightening without FXIs has a negative balance sheet effect on firms with dollar debt, but FXIs can mute this spillover. For this result to be true, it must be the case that a US monetary tightening depreciates local exchange rates when countries do not counter-intervene, but does not depreciate them when they counter-intervene. This is because the local depreciation increases the repayment of US dollar debt in terms of local currency and tightens the balance sheet of firms with US dollar debt.

To check this hypothesis, we estimate Equation 2, where now the dependent variable is replaced with the change in the log of the exchange rate in the country c between dates $t - 1$ and $t + 1$. The exchange rate is defined as the value of one US dollar in terms of local currency, so that higher values imply an appreciation of the US dollar or a depreciation of local currency. $\beta > 0$ implies that a surprise tightening of US monetary policy depreciates the local exchange rate h days after the meeting. We control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, the one-month lagged policy rate, one-year lagged GDP, inflation, the trade balance over GDP ratio, the unemployment rate, and their interaction with the FFR shock.

In Table 4, columns (1) and (2) show the results for countries with and without intervention, respectively. If the local monetary authorities do not intervene, when the Fed unexpectedly hikes interest rates, the domestic currency depreciates in a statistically significant manner (by 2.25% in response to a 10 bp surprise hike). However, if local policymakers counter-intervene in the FX market by selling the US dollar, the exchange rate remains flat around the FOMC meetings. To check that the effect of FXIs is large enough, in column (3), we estimate Equation 4 after replacing the dependent variable with a change in the exchange rate. Without interventions, a 10 bp surprise hike in the Fed funds rate leads to a 2.01% exchange rate depreciation. However, if central banks counter-intervene by selling the US dollar, the depreciation is reduced by 2.02% compared to a scenario without any counter-interventions, and this difference is statistically significant. Thus, interventions offset the exchange rate depreciation caused by the Fed's surprise.

To study how the effect of FXIs accumulates over time, we re-estimate Equation 2 for all $h \in [-5, 5]$. Figure 5, panel (a) plots the estimates of β_h coefficient. Before the FOMC meeting, there is little difference in exchange rates between countries with and without FXIs. However, while countries that do not counter-intervene by selling the US dollar experience persistent depreciation, those that counter-intervene do not experience depreciation. The depreciation builds up slightly over time, consistent with [Roussanov and Wang \(2023\)](#). Moreover, to compare the trends of exchange rates around FOMC meetings in countries with and without FXIs, panel (b) re-estimates the γ_h coefficient in Equation 4 for the exchange rate. $\gamma_h < 0$ for $h > 0$ implies that the exchange rate depreciates less when the central banks counter-intervene against the

Fed compared to a no-intervention case.³⁷ These results suggest that FXIs are successful in stabilizing the exchange rate in response to unexpected monetary shocks, giving further support to the balance sheet stabilization channel of FXIs for firms with dollar debt.

5.4 Expenditure Switching Channel

In the previous sections, we have shown that FXIs benefit firms with dollar debt by muting the negative balance sheet channel of US monetary tightening. However, a US tightening may also have other spillover effects, most importantly via exporting firms. In this section, we first study the effect of US monetary tightening and FXIs on exporters and non-exporters. Next, we test whether exports, as well as international sales and assets, can explain the negative effects of FXIs on firms without dollar debt.

US monetary tightening has two offsetting effects on exporting firms. On the one hand, depreciation induced by contractionary US monetary policy may increase foreigners' demand for exports due to lower relative prices (expenditure switching channel). On the other hand, a US monetary tightening may reduce US demand for goods and thus the demand for exports (a negative demand channel). If FXIs mute the depreciation-induced expenditure switching channel but do not mute the negative demand channel due to intertemporal substitution, we would expect FXIs to be costly for exporters. In addition, since almost all exporting firms in the sample do not have dollar debt (Appendix A), the balance sheet channel is likely muted, and the expenditure switching channel dominates.

To check this possibility, we study the effect of FXIs on exporters and non-exporters. In Figure 6, we conduct a similar exercise to Figure 2 for exporters and non-exporters. Interestingly, although we do find some suggestive evidence for expenditure switching channels, the effect is quantitatively small. Figure 6, panel (a) studies the stock price response to Fed hikes for exporters in countries with and without FXIs, respectively. When the Fed tightens, the stock price of exporters increases without FXIs but decreases with FXIs. However, the magnitude of the stock price decline is small (3.2% at the trough in response to the 10 bp US hike) and the statistical significance is low. In contrast, Figure 2 suggests that stock prices for firms with dollar debt decrease significantly without FXIs (9.0% at the trough), and FXIs mitigate this decline. Next, panel (b) plots the interaction coefficient between FXIs and the export indicator. We show that FXIs decrease the stock price response for exporters compared to cases without FXIs. However, the effect is statistically significant only two days after the FOMC announcement, and the significance disappears five days after the event. In contrast, Figure 2, panel (b) suggests that FXIs have an immediate positive impact on firms with dollar debt on the FOMC announcement date, and the effect is persistent over time. Our estimates suggest that, although FXIs have the

³⁷The effect of FXIs is statistically significant one day after the interventions. This is potentially due to the difference in time zones. Since the United States is one of the most Western countries in the world, the FOMC announcement does not affect the exchange rate in Eastern countries, such as Japan, on the same date. The limitation of our research is that we do not have intra-day data on the exact timing of FXIs in each country.

cost of mitigating expenditure switching channels for firms that export, the benefits for firms with US dollar debt are larger. Ultimately, it may depend on the composition of firms whether FXIs increase or decrease stock prices in the aggregate. For countries in which a large share of firms borrow in US dollars, the positive effect may dominate, but for countries with a large exporting sector, the negative effects could be substantial.

Next, we study what explains the negative effects of FXIs on firms without dollar debt, as shown in Table 1 of Section 5.1. One possibility is the expenditure switching channel for exporting firms. As discussed in Appendix A, most of the exporting firms in our sample do not have dollar debt, so the balance sheet channel is muted for exporting firms. Hence, “Intervening against the Fed” may negatively affect the stock price of firms with exports but without dollar debt. In addition to exports, US monetary policy and FXIs may affect firms with international sales or assets.³⁸ US monetary tightening may benefit these firms by appreciating the US dollar and increasing the value of international sales or assets in terms of the local currency. However, FXIs may harm these firms by appreciating the local currency and mitigating this positive valuation effect.

In Table 5, we test which channels described above can explain the negative effects of FXIs on firms without US dollar debt. Panels (a) and (b) show the results without FXI and with FXI, respectively. If we do not exclude firms with exports, international sales, and international assets (column 1), the coefficient on the FFR_t term is larger with intervention ($\beta = -0.158$, panel b) than without intervention ($\beta = -0.097$, panel a). In columns (2)-(4), we exclude firms with exports, international sales, and international assets, respectively. In column (5), we exclude all of these firms. The table shows that if we exclude firms with international sales or assets, the negative coefficient on the FFR_t term with interventions becomes statistically insignificant (columns 3-5 of panel b) and no longer different from the case without interventions (columns 3-5 of panel a) in a statistically significant manner. This result suggests that the negative effect of interventions on firms without dollar debt can be explained by firms with international sales or assets.

Our results suggest an important policy tradeoff. Intervening against the Fed stabilizes currency risk premia, portfolio flows, and the stock prices of firms, especially those with US dollar debt. However, interventions can also harm firms with international revenues or assets. Whether central banks should counter-intervene or not depends on the relative strengths of these different channels.

We acknowledge that we do not find evidence that FXI benefits firms without dollar debt, which constitute the majority of our sample firms.³⁹ As discussed in Section 3.1, the limitation of our analysis is that it is difficult to obtain firm-level data on the invoicing currency and destination of exports, international sales, and international assets for a panel of multiple

³⁸Exports are defined as sales of domestically produced goods in foreign countries, while international sales are defined as sales of goods in foreign countries, regardless of whether goods are produced domestically or abroad.

³⁹We also do not find evidence that FXI affects the country-level stock market index, which measures the overall stock market performance.

countries. Another limitation is that Worldscope has missing values for exports. In this paper, we focus on the balance sheet heterogeneity since we have detailed firm-level data on the currency composition of balance sheets. We believe that improved data availability in the future would allow for more accurate testing of FXI transmission channels across dimensions of firm heterogeneity.

5.5 Size of Intervention

While we mainly use the $(0, 1, -1)$ categorical variable for FXIs, policymakers may be interested in how much volume of FXIs is sufficient to offset the effect of US monetary policy shocks on the stock market. To address this question, we replace this categorical variable with the volume data of FXI. The FXI volume is measured as a percentage ratio over GDP, and a positive value implies a net purchase of the local currency (net sales of the US dollar). We standardize the FXI volume so that the mean is zero and the standard deviation is unity. We have replaced the categorical variable with the intervention volume data, both in the first- and second-stage regressions.

Table 6 shows that our result is robust to using volume data of interventions instead of a categorical variable. Column (1) shows that a 10 bp increase in the Fed funds rate lowers the stock price of firms with US dollar debt by 1.5 percent, compared to firms without US dollar debt. However, one-standard-deviation US dollar sales (90 million dollars) mitigate this stock price decline by 1.4 percent.⁴⁰ In other words, US dollar sales of 101 million dollars fully offset the stock market transmission of US monetary policy shocks via the balance sheet channel. This result is robust after controlling for the country-time fixed effect (column 2).

We compare this estimate with the actual FXI volumes by central banks. We take the standard deviation of FXI volumes across all unexpected counter-intervention episodes against US monetary shocks. Across all our sample countries, one standard deviation of FXI equals 139 million dollars. If we focus on each country, one standard deviation of FXI equals 114 million dollars in Argentina and 124 million dollars in Brazil. Hence, our estimate is close to the actual volume of counter-interventions by central banks, especially in emerging economies. These results suggest that the volume of FXI is quantitatively meaningful in stabilizing the stock market spillovers of US monetary shocks via balance sheets.

The reason we use a $(0, 1, -1)$ categorical variable throughout the paper is to mitigate the endogeneity. As discussed in [Ito and Yabu \(2007\)](#) and [Menkhoff et al. \(2021\)](#), the decision on whether to intervene or not depends on general financial market and economic conditions in the medium to long run. In contrast, the decision on how much to intervene depends on contemporaneous information on exchange rates or asset prices. We follow the literature and use the categorical variable as a baseline measure to minimize this endogeneity concern.

⁴⁰The intervention of 90 million dollars corresponds to around 0.03% of GDP when evaluated at the median level of GDP across all samples.

5.6 Foreign Exchange Reserves

Central banks' FX reserves have increased significantly over the last decades. Figure A4 shows the volume of FX reserves and the FX reserves over GDP ratio in our sample of countries. In the 1990s, FX reserves were low (roughly 0.21 trillion US dollars), and countries had limited instruments to insulate themselves from US monetary spillovers. However, reserves have increased by a magnitude of around 18 since the 1990s, reaching a peak of 3.91 trillion dollars in 2021. The FX reserve-to-GDP ratio also grew from 4.3% in 1990 to a peak of 32.5% in 2020. So far in 2022, the tightening of US monetary policy has not yet had severe negative consequences for emerging market economies, raising the question of whether FXIs are more effective when countries have relatively large FX reserves, as they can sell off those accumulated reserves to prevent sudden depreciations of exchange rates during times of crisis.

To test this hypothesis, we study the effects of FXIs on exchange rates and stock prices in countries with heterogeneous degrees of FX reserves. We use annual data on total FX reserves in the IMF International Financial Statistics.⁴¹ We define large and small FX reserves if the volume of FX reserves is larger or smaller than the median, respectively.

In Table 7, columns (1) and (2) estimate Equation 4 for exchange rates in countries with large and small reserves, respectively. We find that, in countries with large FX reserves, a 10 bp Fed hike leads to a 2.5% depreciation in the exchange rate, but the depreciation is fully stable when the country intervenes. However, in countries with small FX reserves, the effect of FXIs is reduced to 1.6% and it is statistically less significant. Similarly, columns (3) and (4) estimate Equation 5 for stock prices of firms based in countries with large and small reserves, respectively. An increase in the Fed funds rate leads to a decline in stock prices for firms with dollar debt relative to those without dollar debt. FXIs mitigate this balance-sheet spillover in countries with large FX reserves (4.2pp), but the effect is weaker in countries with small FX reserves (1.3pp). Columns (5) and (6) show this result is robust even after including country-date fixed effects. These results suggest that FX reserve accumulation plays a crucial self-insurance role in mitigating US monetary spillovers.

5.7 Robustness Checks

Appendix A provides various robustness checks to confirm our result, including debt maturity, intensive and extensive margins of dollar debt, alternative specification for unexpected counter-interventions, sterilized interventions, including periods with zero FFR shock, excluding global financial crisis and zero lower bound on the interest rate, international sales and asset holdings, currency denomination of stock price, and currency risk hedging. Furthermore, we control for heterogeneous cross-country characteristics that can confound our estimations,

⁴¹The dataset provides total FX reserves at a country level, but the currency composition of reserves is only available at the world level. [Iancu et al. \(2022\)](#) and [Ito and McCauley \(2020\)](#) provide data on the currency composition of FX reserves. Since the currency composition data is not available for all of our sample countries, we use the total FX reserves data.

including advanced and emerging economies, capital account openness, trade openness, and development. Overall, the key economic implication of the balance sheet channel of FXIs remains robust under these different settings.

6 Conclusion

US monetary policy has significant spillover effects on other countries. In this paper, we investigate those mechanisms of US monetary policy using a high-frequency approach that allows us to understand the effects and channels through which monetary policy affects foreign economies more directly. When countries do not intervene, an unexpected US monetary policy tightening induces foreign exchange rate depreciation against the US dollar, a rise in the currency risk premium, portfolio outflows, and declines in firms' stock prices. The stock prices decline significantly more for firms that borrow in US dollars, mirroring the experience of earlier episodes, such as the 1990s or during the taper tantrum.

In the 1990s, central bank holdings of foreign exchange reserves were low, and they had limited ability to protect themselves against the spillovers of US monetary policy. However, since the 1990s to today, FX reserves have grown by a magnitude of around 16, as shown in Figure A4, likely in anticipation that those reserves can act as a self-insurance mechanism.

In this paper, we have shown that “intervening against the Fed” mitigates exchange rate depreciation, the rise in risk premiums, and portfolio rebalancing. When US monetary policy tightens and central banks counter-intervene, changes in exchange rates, currency risk premiums, and portfolio flows remain stable. Furthermore, interventions mitigate stock price declines for firms, with disproportionately stronger effects for firms with US dollar debt. These results suggest that intervening against the Fed mutes the balance sheet channel of exchange rates triggered by US monetary policy and can protect countries from exposure to the Global Financial Cycle.

Overall, countries today should be less vulnerable to US monetary policy than in previous tightening cycles. However, some countries' reserves remain low, especially those of low-income countries, and their ability to protect themselves from the global financial cycle remains limited.

While we do not study the optimality of the policy interventions, in theory, intervening against the Fed can be optimal if US monetary policy triggers a risk appetite shock ([Basu et al. 2020](#)), pecuniary externalities ([Fanelli and Straub 2021](#)), or local inflationary pressures ([Yago 2024](#)). Further understanding general equilibrium implications and the optimality of policies is an important agenda for future research.

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Table 1: Stock Price: Baseline Regression

(a) Without Intervention

Dependent variable:	$\Delta \text{Stock Price}_{i(c),t}$			
	Dollar Debt	No Dollar Debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock _t	-0.660*** (0.117)	-0.094** (0.045)	-0.097** (0.045)	
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}			-0.314*** (0.087)	-0.259*** (0.071)
R^2	0.093	0.032	0.031	0.083
N	1,926	103,155	105,114	105,114
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

(b) With Intervention

Dependent variable:	$\Delta \text{Stock Price}_{i(c),t}$			
	Dollar Debt	No Dollar Debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock _t	-0.217** (0.105)	-0.149*** (0.056)	-0.158** (0.061)	
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}			-0.001 (0.042)	-0.033 (0.035)
R^2	0.114	0.209	0.194	0.270
N	1,258	9,915	11,178	11,178
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Dollar Debt_{i(c),y-1(t)} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 2: US Monetary Spillover without Interventions (Controlling for Financial Channels)

(a) Firms without US Dollar Debt

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _{<i>t</i>}	-0.094** (0.045)	-0.030 (0.034)	-0.036 (0.044)	-0.093** (0.045)	-0.094** (0.045)	0.009 (0.037)
R^2	0.032	0.044	0.042	0.035	0.032	0.054
N	103,155	103,155	95,668	103,155	103,155	95,668
VIX		✓				✓
Currency Risk Premium			✓			✓
Foreign Yield				✓		✓
Capital Flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

(b) Firms with US Dollar Debt

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _{<i>t</i>}	-0.660*** (0.117)	-0.626*** (0.109)	-0.580*** (0.117)	-0.666*** (0.111)	-0.549*** (0.126)	-0.552*** (0.130)
R^2	0.093	0.104	0.113	0.094	0.100	0.127
N	1,926	1,926	1,724	1,926	1,926	1,724
VIX		✓				✓
Currency Risk Premium			✓			✓
Foreign Yield				✓		✓
Capital Flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_{*t*} is the Fed funds rate shock in basis points. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), we control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We take the sample firms in countries without counter-interventions. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 3: Balance-Sheet Channel: Firms with Dollar Debt (Controlling for Financial Channels)

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _t	-0.647*** (0.116)	-0.561*** (0.100)	-0.600*** (0.120)	-0.652*** (0.112)	-0.558*** (0.131)	-0.487*** (0.127)
FFR Shock _t × Intervention _{c,t}	0.449*** (0.130)	0.386*** (0.144)	0.415*** (0.132)	0.452*** (0.129)	0.360** (0.142)	0.330** (0.152)
R^2	0.091	0.100	0.111	0.092	0.097	0.120
N	3,206	3,206	2,635	3,206	3,206	2,635
VIX		✓				✓
Currency Risk Premium			✓			✓
Foreign Yield				✓		✓
Capital Flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Intervention_{c,t} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), we control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We take the sample firms with dollar debt. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 4: Exchange Rate: Baseline Regression

Dependent Variable:	$\Delta\text{Exchange Rate}_{c,t}$		
	No Intervention	Intervention	Both
	(1)	(2)	(3)
FFR Shock _t	0.225*** (0.069)	0.004 (0.021)	0.201** (0.072)
Intervention _{c,t}			0.266 (0.155)
FFR Shock _t × Intervention _{c,t}			-0.202** (0.072)
R^2	0.108	0.083	0.084
N	418	417	836
Country FE	✓	✓	✓

Note: $\Delta\text{Exchange Rate}_{c,t}$ is the change in the log of the exchange rate from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. FFR Shock_t is the Fed funds rate shock in basis points. Intervention_{c,t} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. We control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, the one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the country and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 5: Stock Price (Excluding Firms with Foreign Revenues and Assets)

(a) Without Intervention

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$				
	(1)	(2)	(3)	(4)	(5)
FFR Shock _t	-0.097** (0.045)	-0.095** (0.044)	-0.089** (0.044)	-0.091* (0.046)	-0.082* (0.043)
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}	-0.314*** (0.087)	-0.316*** (0.088)	-0.292*** (0.082)	-0.313*** (0.088)	-0.313*** (0.085)
R ²	0.031	0.032	0.035	0.035	0.036
N	105,114	96,056	93,608	97,478	83,128
Exclude Firms with Exports		✓			✓
Exclude Firms with International Sales			✓		✓
Exclude Firms with International Assets				✓	✓
Firm FE	✓	✓	✓	✓	✓

(b) With Intervention

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$				
	(1)	(2)	(3)	(4)	(5)
FFR Shock _t	-0.158** (0.061)	-0.157** (0.062)	-0.089 (0.082)	-0.072 (0.049)	-0.085 (0.077)
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}	-0.001 (0.042)	0.002 (0.045)	-0.044 (0.081)	-0.015 (0.059)	-0.041 (0.086)
R ²	0.194	0.186	0.257	0.205	0.246
N	11,178	10,640	6,153	9,713	5,606
Exclude Firms with Exports		✓			✓
Exclude Firms with International Sales			✓		✓
Exclude Firms with International Assets				✓	✓
Firm FE	✓	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Dollar Debt_{i(c),y-1(t)} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. In column (2), we exclude firms with exports. In column (3), we exclude firms with international sales. In column (4), we exclude firms with international assets. In column (5), we exclude firms with exports, international sales, and international assets. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 6: Stock Price: FXI Volume

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$	
	(1)	(2)
FFR Shock _{<i>t</i>}	-0.097** (0.045)	
FFR Shock _{<i>t</i>} $\times$ Intervention _{<i>c,t</i>}	0.005 (0.063)	
FFR Shock _{<i>t</i>} $\times$ Dollar Debt _{<i>i(c),y-1(t)</i>}	-0.152*** (0.044)	-0.167*** (0.046)
FFR Shock _{<i>t</i>} $\times$ Intervention _{<i>c,t</i>} $\times$ Dollar Debt _{<i>i(c),y-1(t)</i>}	0.142** (0.058)	0.111** (0.051)
R^2	0.034	0.086
N	114,448	114,448
Firm FE	✓	✓
Country $\times$ Date FE		✓

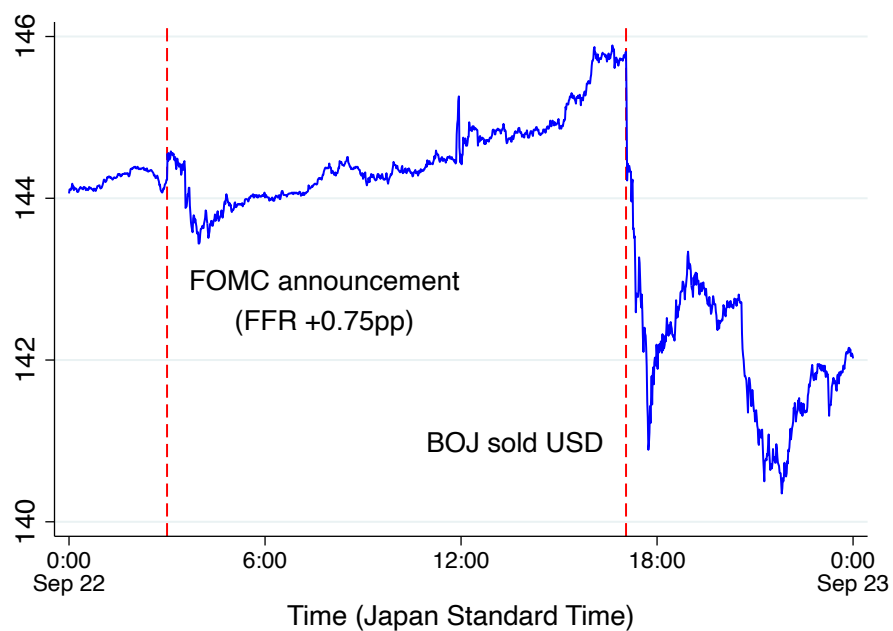
Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_{*t*} is the Fed funds rate shock in basis points. Intervention_{*c,t*} is the standardized volume of unexpected intervention. The FXI volume is measured in terms of percentage ratio over GDP. A positive value indicates a net purchase of the local currency. Dollar Debt_{*i(c),y-1(t)*} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table 7: FX Reserves

Dependent variable:	$\Delta \text{Exchange Rate}_{c,t}$		$\Delta \text{Stock Price}_{i(c),t}$			
FX reserve:	Large	Small	Large	Small	Large	Small
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _{<i>t</i>}	0.252*** (0.065)	0.122 (0.071)	-0.075 (0.067)	-0.126 (0.094)		
FFR Shock _{<i>t</i>} × Intervention _{<i>c,t</i>}	-0.298*** (0.089)	-0.161* (0.080)	-0.108 (0.079)	0.162 (0.143)		
FFR Shock _{<i>t</i>} × Dollar Debt _{<i>i(c),y-1(t)</i>}			-0.387** (0.151)	-0.261*** (0.086)	-0.352*** (0.108)	-0.230*** (0.079)
FFR Shock _{<i>t</i>} × Intervention _{<i>c,t</i>} × Dollar Debt _{<i>i(c),y-1(t)</i>}			0.418** (0.172)	0.137 (0.112)	0.317*** (0.118)	0.201* (0.102)
<i>R</i> ²	0.121	0.149	0.053	0.051	0.109	0.098
N	422	413	90,860	25,880	90,860	25,880
Country FE	✓	✓				
Firm FE			✓	✓	✓	✓
Country × Date FE					✓	✓

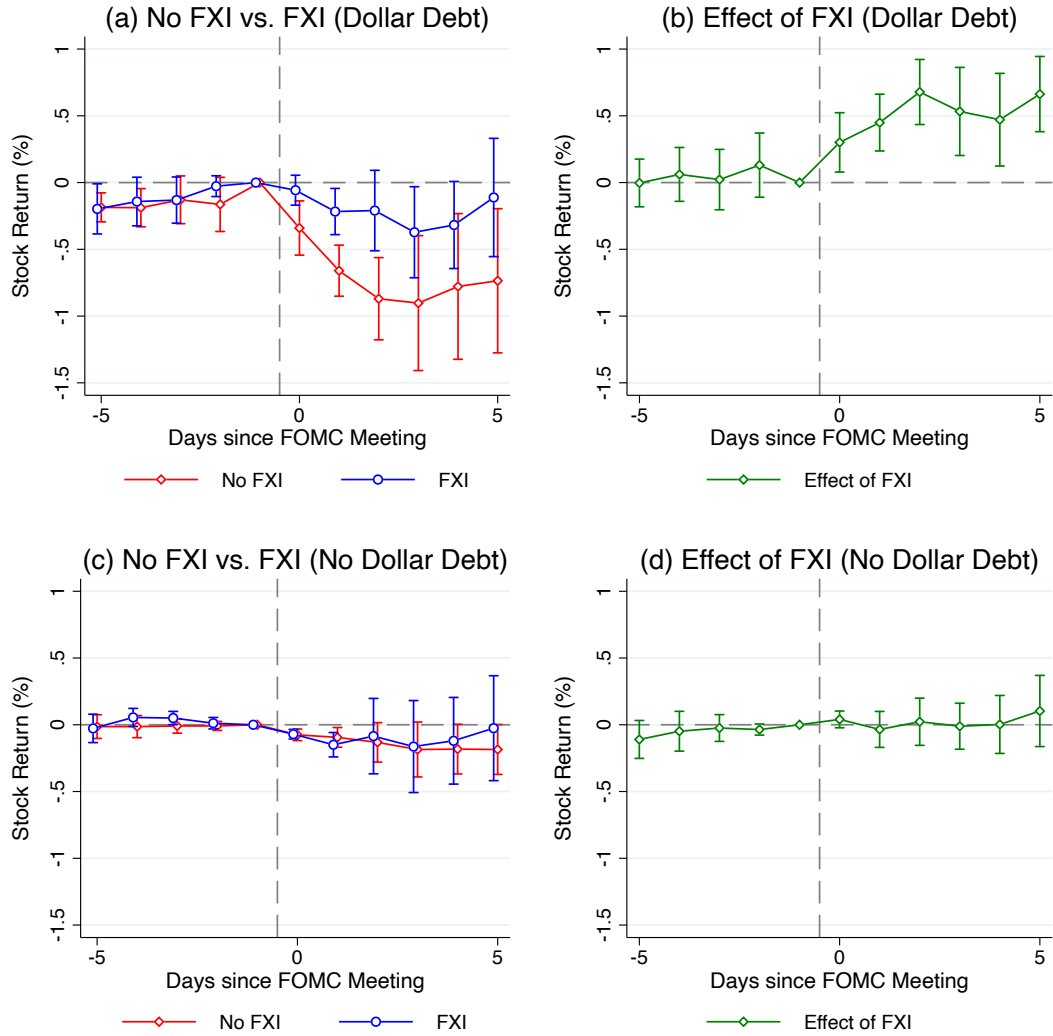
Note: Columns (1) and (2) show the results for country-level exchange rates, and columns (3) to (6) show the results for firm-level stock prices. $\Delta \text{Exchange Rate}_{c,t}$ is the change in the log of the exchange rate from date $t - 1$ to $t + 1$, and $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_{*t*} is the Fed funds rate shock in basis points. Intervention_{*c,t*} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt_{*i(c),y-1(t)*} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. Large and small FX reserves are defined so that the volume of FX reserves is larger and smaller than the median, respectively. In columns (1) and (2), we control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. In columns (3)-(6), we control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Figure 1: Spot Exchange Rate: US Dollar to Japanese Yen



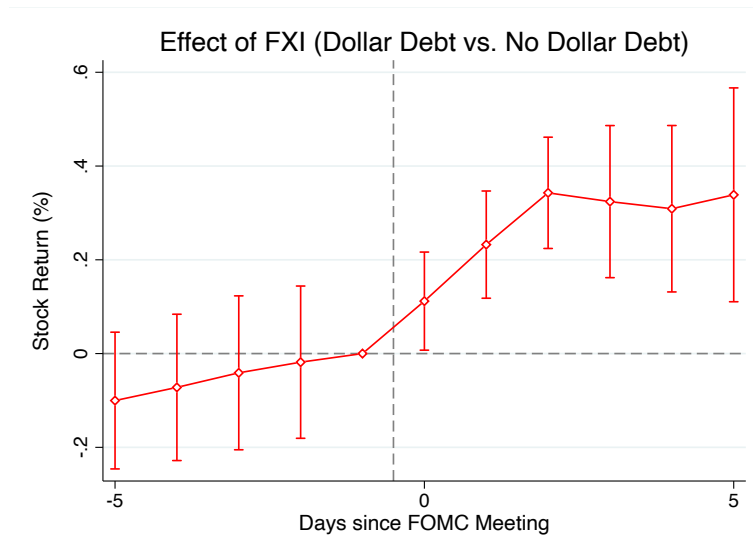
Note: The figure reports the minute-by-minute US dollar to Japanese yen spot exchange rate on September 22, 2022. The exchange rate is defined as the value of one US dollar in terms of yen, and a higher value implies the appreciation of the dollar or depreciation of the Japanese yen. Source: Datastream.

Figure 2: Stock Price: Difference-in-Difference



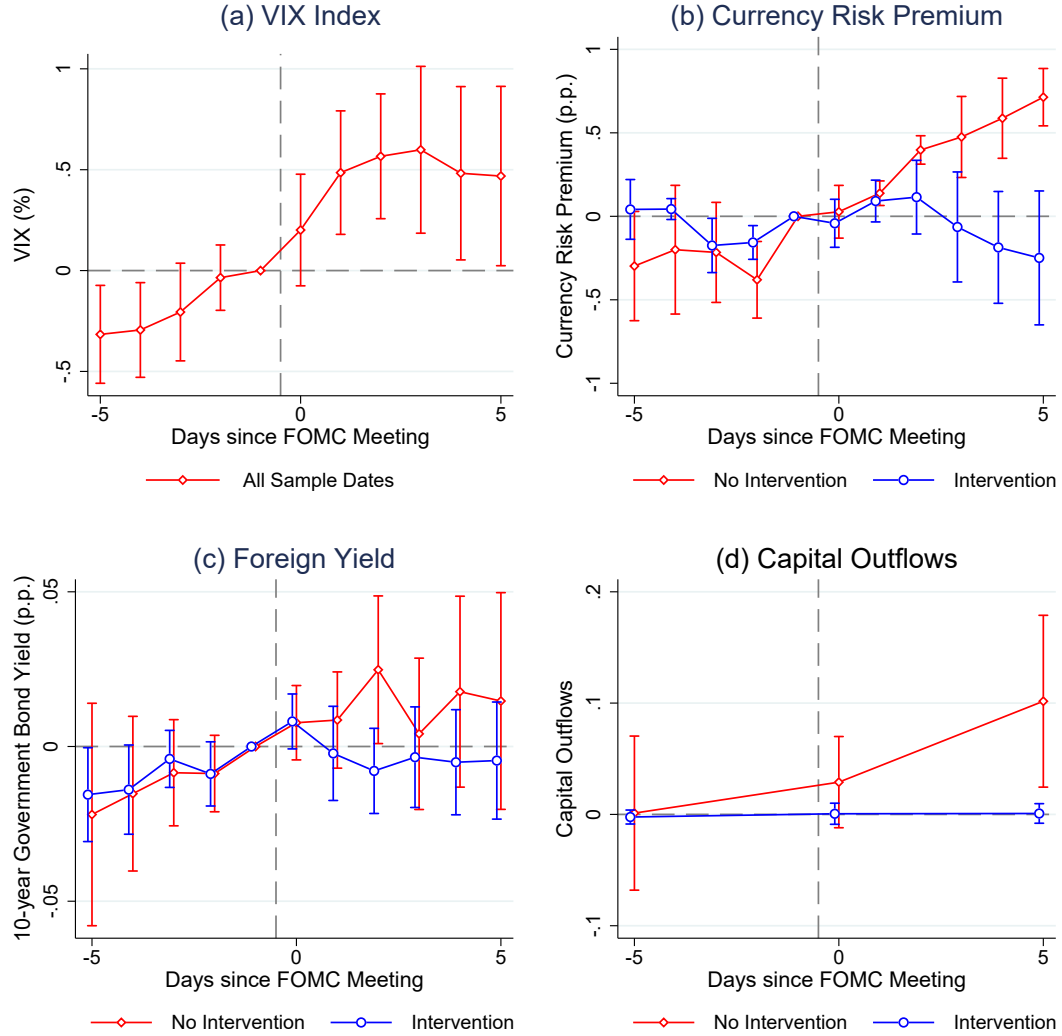
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Equations 2 and 4 for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

Figure 3: Stock Price: Triple Interaction



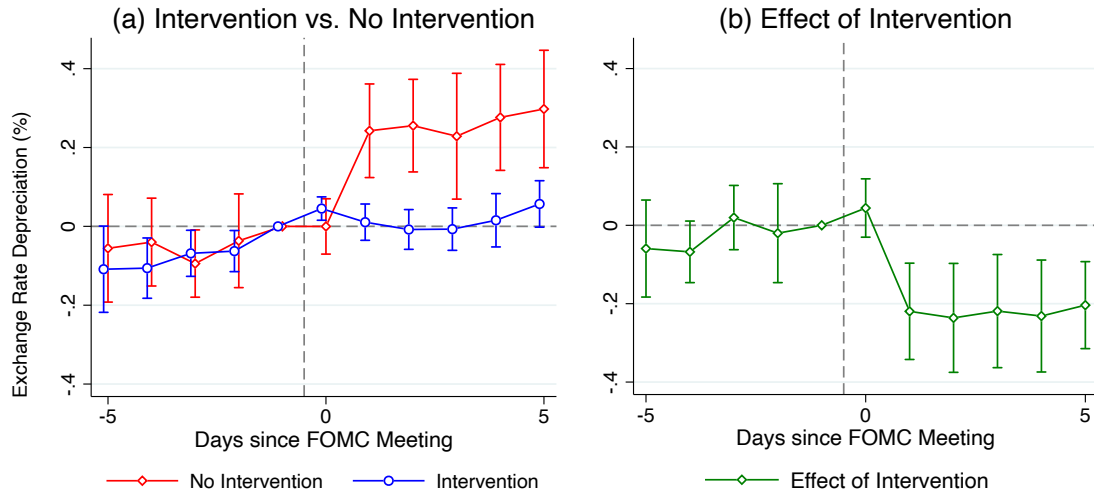
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Equation 5 for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

Figure 4: Different Transmission Channels of FXI



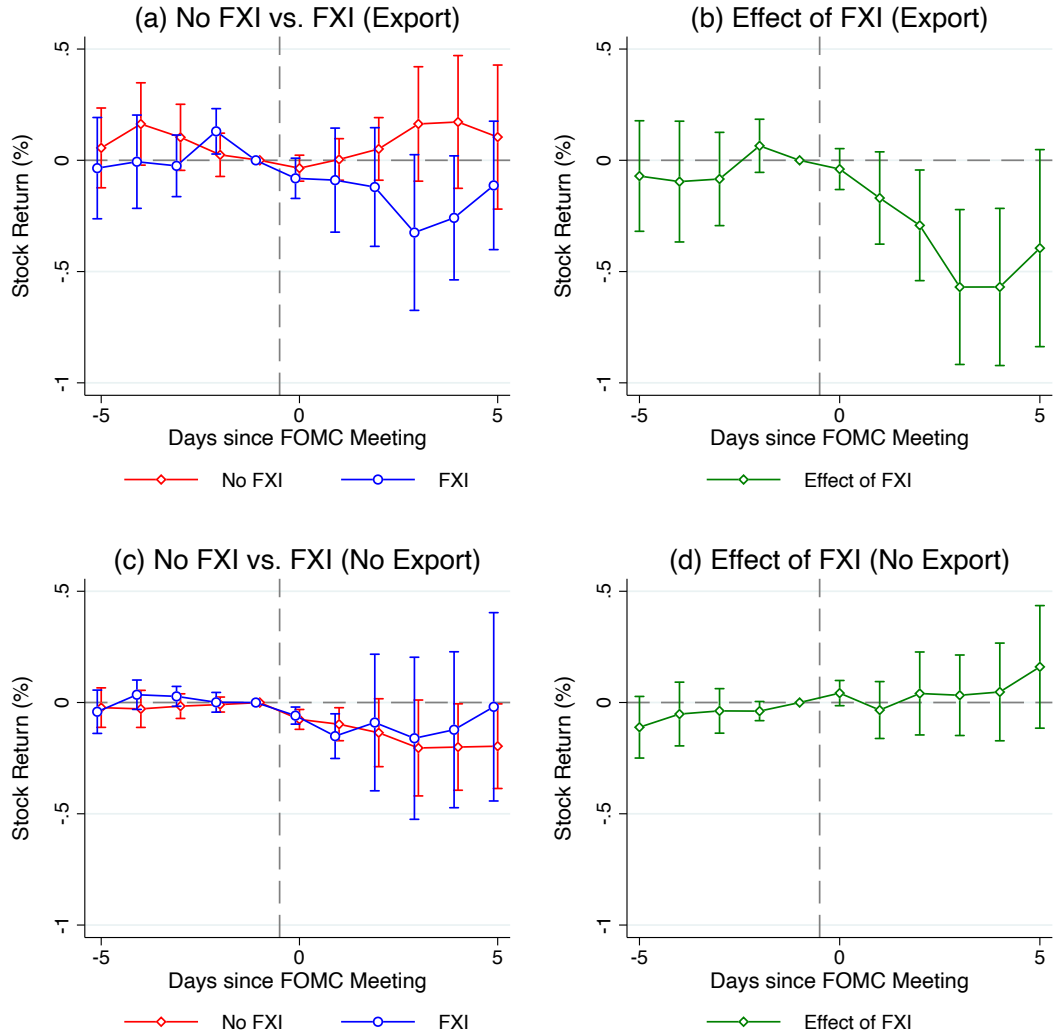
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on the VIX index, currency risk premium, foreign bond yield, and equity capital flows. In panel (a), the dependent variable is the change in the 3-month US VIX index. In panel (b), the dependent variable is the change in currency risk premium. A positive value implies a higher excess return on the foreign currency over the US dollar. In panel (c), the dependent variable is the change in foreign 10-year government bond yields. In panel (d), the dependent variable is the equity capital flows. A positive value implies a capital outflow from foreign countries. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level (panel b) and clustered at the date level (panels a, c, and d). Note that, given low clusters due to data availability, we cannot double cluster in panels (c) and (d). The confidence interval is 90%.

Figure 5: Exchange Rate: Difference-in-Differences



Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on exchange rates. See Equation 4 for the exact specification. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level. The confidence interval is 90%.

Figure 6: Stock Price: Expenditure Switching Channel



Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices for exporters and non-exporters. See Equations 3 and 5 for the exact specifications. We control for one-year lagged dollar debt, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.

Internet Appendix

A Robustness Checks

This section provides robustness checks to test whether our results are robust and confirm the balance sheet channel of FXI. In Tables A6, A7, and A8, we reestimate Equation 5 for stock prices under various alternative settings. The negative coefficient on $FFR_t \times USD_{i(c),y-1(t)}$ implies a balance sheet channel of US monetary policy shocks, while the positive coefficient on $FXI_{c,t} \times FFR_t \times USD_{i(c),y-1(t)}$ implies that FXIs can mute the balance sheet channel. Similarly, in Table A9, we reestimate Equation 4 after replacing the dependent variable with exchange rate depreciation. The positive coefficient on FFR_t implies that the US tightening depreciates the local exchange rate, while the negative coefficient on $FFR_t \times FXI_{c,t}$ implies that FXIs mitigate this depreciation. Overall, the results remain qualitatively the same in various settings.

A.1 Debt Maturity

To further refine our identification strategy, we use the firms' debt maturity structure, which is arguably exogenous to both FXIs and monetary policy shocks. If firms happen to have dollar debt whose maturity is around unexpected Fed hikes, the cost of rolling over debt increases. However, if the debt does not mature around the FOMC events, the effect on rollover costs is negligible. Hence, FXIs that counteract Fed hikes should disproportionately benefit firms with dollar debt that matures around the FOMC announcement dates. Since the maturity structure is exogenous to exchange rate movement, the stock price response can potentially be interpreted causally.

To test this hypothesis, we divide the dollar debt into maturing and non-maturing dollar debt. Maturing dollar debt is defined as the debt whose repayment currency is the US dollar and which matures two quarters (six months) before or after the FOMC announcement date. Non-maturing dollar debt is defined as dollar debt that does not mature within a one-year window around the announcement. We redefine the dummy for dollar debt so that it takes one if firms have maturing dollar debt and zero if firms do not.

Table A6 shows the result. Column (1) takes the dummy for dollar debt regardless of maturity, as in column (3) of Table A4, and column (2) takes the dummy for maturity dollar debt as defined above. Column (1) shows that, without FXIs, a 10bp increase in the Fed funds rate leads to a 3.1pp decline in stock prices for firms with dollar debt (compared to those without dollar debt), while FXIs mitigate this decline by 3.2pp. However, column (2) shows that, for firms with maturing dollar debt, the same increase in the Fed funds rate leads to a 5.5pp decline in stock prices without FXIs and a 5.4% mitigation with FXIs, both of which are greater than the benchmark case. This suggests that the balance sheet channel of US monetary policy and FXIs is driven by the increased cost of debt rollover around the maturity date. For robustness,

columns (3) and (4) redefine maturity dollar debt as the dollar debt that matures three quarters or one year around FOMC announcements and obtain a similar result.

A.2 Intensive and Extensive Margins of dollar debt

The baseline result on the balance sheet channel focuses on whether firms borrow in US dollars (extensive margin). However, in reality, the effect of interventions may also depend on how much they borrow in dollars (intensive margin). Firms with large amounts of dollar debt would be more exposed to currency risk than those with small amounts of dollar debt. To test this intensive margin, we define an indicator variable for “low dollar debt,” which takes one if the firms’ share of dollar debt over total debt is larger than the 25th percentile (conditional on having a positive amount of dollar debt) and zero otherwise (including firms with dollar debt lower than the 25th percentile and firms without dollar debt).⁴² Similarly, we define dummies for “high dollar debt” for firms with a median dollar debt share, and “all dollar debt” for firms that only issue dollar debt. In our sample, around 29% of our firm-date observations are driven by firms only with dollar debt. This definition can capture the right tail of the distribution of the dollar debt share across firms.

Table A7 shows the result. Comparing columns (1)-(3), we find that both the balance sheet channel of US monetary policy shocks (negative coefficient γ on $FFR_t \times USD_{i(c),y-1(t)}$) and FXI’s mitigation of monetary spillovers (positive coefficient θ on $FXI_{c,t} \times FFR_t \times USD_{i(c),y-1(t)}$) are the strongest for firms with high dollar debt ($\gamma = -0.345$, $\theta = 0.468$) and the weakest for those with low dollar debt ($\gamma = -0.303$, $\theta = 0.293$).

Moreover, instead of a dummy variable for large dollar debt, we take the continuous share of dollar debt over total debt. The dollar debt share is standardized so that one unit of dollar debt share corresponds to one standard deviation (around 12.2%). Column (4) shows that when the share of dollar debt is higher by one standard deviation, a 10bp increase in the Fed funds rate leads to a 0.43% decline in stock price without FXI. However, when the central bank intervenes, this decline is mitigated by 0.44%, entirely offsetting the effect of the US monetary shock. These results imply that US monetary shocks and FXIs have a large effect, especially on firms with large dollar debt.

A.3 Alternative Definitions for Unexpected Counter-Intervention

We study different criteria for defining unexpected counteracting interventions. In the benchmark case, counteracting interventions are defined so that, if the Fed funds rate increases on date t , central banks sell the US dollar at least once and never buy the US dollar between dates t and $t + 5$, and vice versa when the Fed funds rate decreases. Then, we estimate Equation 1 and define unexpected counter-interventions if the residual is larger than its median in absolute

⁴²We also check that the result is robust with a continuous measure of dollar debt over total debt.

value. While we use this definition in the benchmark results to simplify the interpretation, we also try two alternative definitions for an unexpected counter-intervention.

First, we define counter-interventions as occurring when the central banks' average net sales of US dollars between dates t and $t + 5$ are positive in case the Fed funds rate increases on date t , and vice versa when the Fed funds rate decreases. Second, counter-interventions are defined as unexpected if the residual from estimating Equation 1 is larger than the 75th percentile rather than the median in absolute value. This removes the estimation bias when using a 0-1 dummy on the left-hand side of Equation 1. Columns (1) and (2) of Tables A8 and A9 show the results under these four alternative definitions. Comparing these findings with column (3) of Tables 1 and 4, we confirm our results on stock prices and exchange rates to be robust. Without FXI, a US monetary tightening shock depreciates the exchange rate and reduces the stock price for firms with dollar debt, but FXIs mitigate this effect.

A.4 Sterilized Intervention

We further investigate the effect of sterilized intervention. Sterilization implies that central banks conduct an open market operation to offset the effect of FXIs on the domestic financial market. We control for daily monetary policy rates to separate the effect of monetary policy and focus on the FX purchases or sales by central banks.⁴³ In the benchmark result, we use monthly data on policy rates since the monthly data are available for all sample countries, but daily data are not. In this section, we instead use the daily policy rate in BIS statistics for a subsample of countries where the data is available. We control for daily policy rates in the first-step and second-step regressions. Column (3) of Tables A8 and A9 shows that our result is robust after controlling for daily policy rates.

A.5 FOMC Event Dates with Zero FFR Shocks

In the baseline analysis, we used 90 FOMC event dates with nonzero changes in the Fed funds rate, since our main focus is the counteracting interventions against unexpected changes in US monetary policy. Here, we also include 46 additional FOMC event dates with zero changes in the Fed funds rate. Column (4) of Tables A8 and A9 shows that our result is robust even after including these dates.

A.6 Excluding Special Periods

We check the possibility that our result is affected by a particular period. Column (5) of Tables A8 and A9 excludes the global financial crisis (between 2007 and 2009), and column (6)

⁴³We do not have information on whether the interventions are sterilized or unsterilized. As discussed in [Fratzscher et al. \(2023\)](#), given that FXIs are a distinct policy tool from the interest rate, it is reasonable to assume that interventions in our sample are generally sterilized.

excludes the period with a zero lower bound on the interest rate (policy rate is below or equal to zero), and our result is robust.

A.7 Market Beta

We take into account the possibility that the effect of the FOMC announcement on stock prices is driven by risk premia as opposed to the exchange rate. To measure the standard risk factor, we define the market beta as $\beta_m = \text{cov}(R_{i(c)}, R_c) / \text{var}(R_c)$, where $R_{i(c)}$ is the stock return of firm i located in country c and R_c is the aggregate stock market index in country c . In Table A8, column (7), we control for the market beta and its interaction with Fed funds rate shock, and the result is robust.

A.8 Controlling for International Sales and Asset Holdings

In the baseline results on stock prices, we control for exports and imports because they affect firms' foreign currency revenues and costs. Other factors potentially affect firms' foreign currency revenue, including international sales and international assets, as firms with international sales or assets may benefit from a domestic currency depreciation. To take this into account, we control for the one-year lagged international sales over the total sales ratio and the international assets over the total asset ratio. The data is available in Worldscope. Table A8, column (8) shows that the result is robust after including these additional controls.

A.9 Currency Denomination of Stock Prices

In the baseline analysis, the firm's stock price is denominated in local currency. In this section, we denominate the stock price in US dollars so that changes in exchange rates affect the valuation of stock prices. When the Fed funds rate increases, not only do stock prices decrease for firms with dollar debt, but the exchange rate depreciates against the dollar as well. Hence, we expect that the stock price in terms of the US dollar decreases more than the one denominated in local currency. This implies that firms with dollar debt are riskier investment opportunities for US international investors compared to local investors. Table A8, column (9) shows that when the stock price is denominated in the US dollar, a 10bp Fed tightening without FXIs decreases the relative stock price for firms with dollar debt by 3.8bp, while FXIs mitigate this decline by 3.8bp. Those effects are greater than the benchmark case in Table 1, column (3), where the stock price is denominated in local currency.

A.10 Currency Risk Hedging

In this section, we address the possibility that firms hedge their exchange rate risk to limit their exchange rate exposure. Unfortunately, detailed firm-level data on derivatives is not

available. Hence, we approximate whether firms hedge their currency risk through a statistical approach. If firms have unhedged US dollar debt, a depreciation of the domestic currency against the US dollar should decrease firms' future profits in domestic currency due to higher debt repayments. In contrast, if firms are fully hedged, either naturally due to revenues in US dollars or financially through derivative positions, a depreciation should not affect their future profits, as the derivative positions or revenues increase proportionately with the depreciation of the domestic currency. To measure the degree of risk hedging, we, therefore, take the correlation between firms' stock prices, accounting for future discounted profits, and the exchange rate over the sample period. A higher value of the exchange rate implies a local currency depreciation, so a positive (negative) correlation implies that firms' profits increase (decrease) when the local currency depreciates. We define hedged and unhedged firms as those whose stock price has a positive and negative correlation with the exchange rate, respectively. Table A10 shows the number of hedged and unhedged firms. Among 261 firms with dollar debt (row a), 64 firms (25%) have a positive correlation and 195 firms (75%) have a negative correlation, indicating that the majority of firms with dollar debt are unhedged against exchange rate risks. Among firms without US dollar debt, one would not expect the firms to have a strong negative correlation with a depreciation of the exchange rate, as debt repayments are not mechanically increasing with the depreciation. In fact, without US dollar debt, a depreciation may even increase profits through an expenditure switching channel. Among 3799 firms without dollar debt (row b), 2409 firms (63%) have a positive correlation with the depreciation, while 1378 firms (36%) have a negative one.

One might also expect that if firms with dollar debt are exporters, they are naturally hedged in terms of cash flows since a depreciation increases export revenues in terms of local currency. This is, in fact, true in our data. Table A10, row (c) shows that, out of 501 exporters, 462 out of 501 exporters (92%) are hedged. Among the firms with dollar debt, only four are exporters. Hence, our sample firms are unlikely to be naturally hedged against exchange rate depreciations, and our result on the dollar debt channel is mainly driven by firms without corresponding export revenues. Our results on FXIs and the stock market are robust even after controlling for exports and imports, as well as their interaction with Fed funds rate shocks. As a robustness check, Appendix A also controls for international sales and asset holdings, and our main results do not change.

To check that our result is not driven by hedged firms, Table A11, column (1) re-estimates Equation 5 after excluding the hedged firms defined above. Column (2) excludes exporters, and column (3) excludes both hedged firms and exporters. Comparing the result with the benchmark case in Table 1, column (3), we find that our result is robust after eliminating the effect of hedging.

The limitation of our analysis is that we do not have data on firms' derivative use. For firms that perfectly hedge their currency risk, we would expect the stock price to be unaffected around FOMC announcements, similar to those firms with no US dollar debt. However, our

result suggests that, without FXIs, the stock price of firms with dollar debt rather decreases. This suggests that our results are lower-bound estimates of the effect of FXIs since controlling for hedging would further strengthen the negative stock price spillover of US monetary shocks. Moreover, the effects would likely be small, as only a very small share of firms use derivatives. For instance, [Casas et al. \(2022\)](#) show that only 2.9% of Colombian firms use FX derivatives. We expect that increased data availability on hedging would further improve our estimates, which is left for future research.

A.11 Cross-Country Heterogeneity

This section explores different dimensions of cross-country heterogeneity. The limitation of our analysis is that, since our country sample is rather small, it is difficult to exploit even more heterogeneity due to power issues. Especially, we have a limited sample of advanced economies with available daily FX interventions data, and thus, the number of unexpected counter-interventions is limited in advanced economies. Hence, estimating the regression separately for advanced economies suffers from the power issue. An alternative to splitting the data is to test whether certain country groups behave differentially by estimating an additional interaction term, which preserves the same number of observations. We explore whether the effect of FX intervention is different across emerging and advanced economies and countries with different degrees of capital controls. In addition, we also study whether different trade openness and development can confound the effects of interventions.

First, we construct an indicator variable, *EME*, which takes one for emerging economies and zero for advanced economies to measure the heterogeneous effects of FX interventions on emerging and advanced economies. Next, to measure the degree of capital account openness, we use both a de jure measure, which captures the legal restrictions on capital flows, and a de facto measure, which captures the actual volume of capital flows. For the de jure measure, we use the capital account openness index, *KAOPEN*, developed by [Chinn and Ito \(2006\)](#). *KAOPEN* is the first principal component of the four indicator variables on capital account restrictions from the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER).⁴⁴ A higher value implies greater capital openness, and a lower value implies strict capital controls or restrictions on international financial transactions. For the de facto measure, we use the External Wealth of Nations database constructed by [Lane and Milesi-Ferretti \(2018\)](#). We measure the capital account openness by taking the sum of gross external assets and liabilities, expressed as a percentage ratio of GDP.

To measure the degree of trade openness, we take the sum of gross exports and imports (as a ratio of GDP). Finally, we control for the GDP per capita to control for the development of

⁴⁴These indicator variables are defined as follows: k_1 is the indicator for multiple exchange rates, k_2 is the indicator for restrictions on the current account, k_3 is the indicator for restrictions on the capital account transactions, and k_4 is the indicator for the requirement of the surrender of export proceeds. *KAOPEN* is a continuous variable, and its mean is zero by construction.

the country. The data on export, import, and GDP are available in the World Bank database. We standardized the capital account and trade openness indices and the GDP per capita so that the mean is zero and the standard deviation is one. The variables are lagged by one year.

To explore the possibility that the effect of FX interventions can be confounded by these variables, we estimate the following quadruple equation:

$$\begin{aligned}
y_{i(c),t+1} - y_{i(c),t-1} = & \lambda_h \text{FFR}_t \times \text{FXI}_{c,t} \times \text{USD}_{i(c),y-1(t)} \times \text{Characteristic}_{c,y-1(t)} \\
& + \theta \text{FFR}_t \times \text{USD}_{i(c),y-1(t)} \times \text{FXI}_{c,t} \\
& + \kappa \text{FFR}_t \times \text{USD}_{i(c),y-1(t)} \times \text{Characteristic}_{c,y-1(t)} \\
& + \gamma \text{FFR}_t \times \text{USD}_{i(c),y-1(t)} \\
& + \mathbf{X}\delta_x + \alpha_{i(c)} + \epsilon_{i(c),t},
\end{aligned}$$

$\text{Characteristic}_{c,y-1(t)}$ is the set of country-level variables as described above, including the indicator for emerging economies, capital openness (de jure or de facto), trade openness, and GDP per capita. The quadruple interaction coefficient λ_h measures differential effects of FXI across countries with heterogeneous characteristics.

Table A12 compares the results when we do not exploit additional heterogeneity (column 1) and when we do (columns 2-6). Column (2) tests whether emerging economies behave differentially. Columns (3) and (4) exploit heterogeneity across the de jure and de facto measures of capital account openness as defined above, respectively. Column (5) uses the trade openness measure. Column (6) exploits heterogeneity across GDP per capita.

The table shows that the baseline coefficients are similar to our baseline: a surprise increase in the Fed funds rate reduces stock prices for firms with dollar debt, and FX interventions mitigate this decline. The quadruple interaction coefficients of these factors with the Fed funds rate shock, FX intervention, and dollar debt are statistically insignificant. Taken at face value, these results would imply that our key mechanism—the balance sheet channel of US monetary shocks and FXIs—is similar across different types of countries, with no obvious confounding factor driving our results. However, we take those results with a grain of salt. The low number of country observations makes it less likely for us to find significant heterogeneity. We hope that in the future, daily FXI data will become more readily available for a larger set of countries, and this exercise can be conducted in a more reliable way.

A.12 Other Robustness Tests

We conduct several other robustness tests. First, to take into account the difference in the number of firms with dollar debt across countries, we weight the regression by the number of firms with dollar debt in each country. In Figure A5, we reproduce Figure 2, panels (a) and (b) after weighting the regression by $1/n_{i(c)}$, where $n_{i(c)}$ is the number of firms with dollar debt in country c where firm i belongs. The result is qualitatively similar to the benchmark case.

Next, to address the concern that our result is driven by one particular country, we exclude each sample country from the regression and show that the result is robust. Each line in Figure A6, panels (a) and (b) show the estimates of Equations 2 after excluding each country from our sample. Without FXI, a Fed funds hike leads to a decline in stock price for firms with dollar debt (panel a), while the stock price remains stable with FXIs (panel b). Next, Figure A6, panel (c) shows the estimates of Equations 4 after excluding each country from our sample. We learn that there is a positive difference in stock price response between the cases without and with FXI. These results are consistent with the benchmark case in Figure 2. Figure A7 conducts a similar exercise for exchange rate. Panels (a) and (b) show that the exchange rate depreciates after a Fed funds hike without FXI, while it remains stable with FXI. Figure A7, panel (c) shows the difference in exchange rate responses without and with FXI. These results are consistent with the benchmark case in Figure 5.

B Construction of the Currency Risk Premium

This section provides a formal definition of currency risk premium. Let $s_{c,t}$ and $f_{c,t}$ be the spot and forward exchange rates in units of currency c against the US dollar at date t . An increase in s implies an appreciation of currency c . Let $i_{c,t}$ and i_t be the 1-month nominal interest rates in country c and the United States at date t .

Following [Lustig et al. \(2011\)](#), we define the 1-month currency risk premium as the interest rate differential minus the exchange rate depreciation:

$$rx_{c,t+30} = i_{c,t} - i_t - \Delta s_{c,t+30}.$$

A positive value implies the foreign currency has an excess return over the US dollar. When the covered interest rate parity condition holds, the forward discount is equal to the interest rate differential:

$$f_t - s_t \approx i_{c,t} - i_t.$$

[Akram et al. \(2008\)](#) find that the CIP condition holds at daily and lower frequencies. Hence, the currency risk premium can also be expressed as the excess return on buying the foreign currency in the forward market and selling it in the spot market:

$$rx_{c,t+30} = f_t - s_{c,t+30}.$$

C Construction of Firm-Level Data

This section provides the definition and cleaning procedure of firm fundamentals and balance sheets. The firm-level data on fundamentals and balance sheets, the variable definitions, and cleaning procedures follow the standard literature on monetary policy and corporate balance sheet risk, for example, [Ottonello and Winberry \(2020\)](#). The cleaning procedure of Capital IQ data follows [Kim et al. \(2020\)](#).

C.1 Data Definition

- *Leverage*: the ratio of total debt over total assets.
- *Share of dollar debt*: the ratio of the total due amount of dollar debt (the sum of due amounts of debt instruments whose repayment currency is the U.S. dollar) over the total due amount of debt denominated in all currencies.
- *Size*: total assets, which are denominated in local currency and deflated by the consumer price index (CPI).
- *Liquidity*: the ratio of cash and short-term investments over total assets.
- *Age*: years after the incorporation date.
- *Export intensity*: the ratio of exports over total sales.
- *Import content of production*: the import content of exports, defined as the contribution of imports to the production of goods and services.

C.2 Data Cleaning Procedure

We only use data from publicly listed firms, as the data on stock prices is available. The sample only includes ultimate corporate parents, which are headquartered in each country. Moreover, the sample excludes the following:

- Firm-year observations in which balance sheet information is not reported.
- Firm-year observations in which the currency composition of debt is not reported.
- Each control, including total assets, principal due, tangible assets, liquidity, and long-term investment, belongs to the top or bottom 1% in each country.
- Leverage belongs to the top 1% in each country.
- Financial firms (SIC industry code: 6000-6999).
- Government institutions.

- Firm-year observations in which the sum of cash and cash equivalents and tangible assets is greater than the total assets.
- Firm-year observations in which the difference between the total assets and the sum of total liabilities and equity is greater than 10,000 U.S. dollars.
- Firm-year observations in which the difference between the sum of principal dues of all individual debt instruments, which is available in the detailed financial statement, and the total principal due of debt, which is available in the main financial statement, is greater than 100,000 U.S. dollars.
- Firm-year observations in which the sum of due amounts of U.S. dollar debt in the detailed financial statement is greater than the total due amounts in the main financial statement.

D Lower-frequency Effect of FXI

This section studies how long the effects of US monetary shocks and FXIs on the exchange rate and stock market last. In Figure A8, we reestimate Equation 2 over h trading days before and after the FOMC announcement, where $h \in [-5, 10]$ (10 trading days equal two weeks). In panel (a), we take the exchange rate as the dependent variable. Panel (a-1) shows that without intervention, a US monetary tightening shock depreciates the foreign exchange rate for 9 days (the red line), while the effect on the exchange rate is limited with intervention (the blue line). Panel (a-2) takes the difference between intervention and no-intervention cases and shows that interventions mitigate the depreciation over five days. Moreover, to study the persistence of financial channels of US monetary shocks and FXI, in panels (b), (c), and (d), we replace the dependent variables with the currency risk premium, foreign bond yield, and capital outflows. We learn that US monetary shocks and FXI affect the currency risk premium over seven days (panel b) and capital flows over 5 days (panel d), while the effect on foreign bond yields is limited. We also learn that, as we take a longer time horizon, the standard error increases since our estimates are affected by unobserved confounders.

Next, to test the persistence of the stock market effects, in Figure A9, we regress Equation 5 over h weeks before and after the FOMC announcement, where $h \in [-2, 10]$. Figure A9, panel (a) plots the coefficient γ_h on $FFR_t \times USD_{i(c),y-1(t)}$ and panel (b) the coefficient θ_h on $FFR_t \times USD_{i(c),y-1(t)} \times FXI_{c,t}$, respectively. Interestingly, the effects of US monetary shocks and FXIs on the stock market identified in the high-frequency data are more persistent and hold in the lower frequency, in contrast to the effect on the exchange rate. Panel (a) shows that a 10 basis point increase in the Fed funds rate decreases the stock price of firms with dollar debt by around 0.5 percent relative to those without dollar debt, and the effect is persistent over 6 weeks (around 1.5 months). Panel (b) shows that FXIs mitigate the stock price decline by around 0.5 to 1 percent. The effect accumulates over time and is persistent over 8 weeks (around 2 months).

We expect that the reason why the estimated effects on the stock market are more persistent is that we can better identify the direct effects of US monetary shocks and FXIs. By exploiting granular firm-level stock price data and including a country-time fixed effect, we can control for time-varying observed and unobserved country-level characteristics. The cleanly identified monetary and FX policy shocks lead to more persistent effects on the stock market. However, we still observe that the standard error increases with a longer time horizon due to unobservable characteristics that affect each firm in our sample, and since the sample gets smaller once we split it into countries with and without interventions. The whole point of this paper is to identify the effects of monetary policy and FXI in a narrow time window of five days to minimize the endogeneity concerns. Studying the lower-frequency effects of FXIs on the exchange rate, stock prices, and the macroeconomy is an important area for future research.

Table A1: Summary Statistics: FFR shock, exchange rate, and stock price

	Mean	Med	S.D.	p5	p95	Obs
(1) FFR shock (basis point)	0.015	-0.48	1.81	-3.1	3.75	90
(2) Exchange rate (% change, $\log(e_{c,t+1}) - \log(e_{c,t-1})$)	0.04	0	0.72	-1.37	1.29	875
(3) Stock price (% change, $\log(p_{i,t+1}) - \log(p_{i,t-1})$)	0.02	0	3.48	-5.61	5.71	124,559

Note: t is the FOMC announcement date. $e_{c,t+1}$ is the exchange rate in country c at date $t + 1$. A higher $e_{c,t+1}$ implies the appreciation of the US dollar or depreciation of the local currency. $p_{i,t+1}$ is the stock price of firm i at date $t + 1$. The stock price is in terms of local currency. Observations (column 6) are the number of FOMC announcement dates (row 1), country $\times$ FOMC announcement dates (row 2), and firm $\times$ FOMC announcement dates (row 3).

Table A2: Interventions around 90 FOMC event dates in sample

Country	Frequency			Volume (Millions USD)		Periods
	Buy USD	Sell USD	Counter	Mean	Median	
	(1)	(2)	(3)	(4)	(5)	(6)
Argentina	59	45	15	86	52	2003-2019
Australia	0	2	2	19	19	2000-2019
Brazil	11	1	8	165	114	2009-2019
Chile	6	0	4	0.091	0.096	2008-2019
Colombia	34	2	18	19	17	2000-2019
Costa Rica	34	32	3	12	8.1	2006-2019
Georgia	9	12	15	3.9	3.1	2009-2019
Hong Kong	83	58	13	70	12	2000-2019
Japan	4	0	1	1115	1493	2000-2019
Mexico	0	24	7	27	22	2000-2011
Peru	72	51	26	23	4.3	2000-2019
Switzerland	0	0	0	0	0	2000-2001
Turkey	1	1	0	5.9	5.9	2002-2019
Total	312	229	111	57	17	2000-2019

Note: Columns (1) and (2) show the frequencies of buying and selling interventions. “Buying interventions” are defined as occurring when central banks buy the US dollar at least once between date t and $t + 5$, where t is the FOMC announcement date; “selling interventions” are defined analogously. Column (3) shows the frequency of counter-interventions, defined as occurring when the central bank sells the US dollar at least once and never buys the currency again between the dates t and $t + 5$ after the Fed funds rate increases at date t ; vice versa when the Fed funds rate decreases. Columns (4) and (5) show the mean and median intervention volumes in terms of millions of US dollars between date t and $t + 5$ over all FOMC announcement dates t , respectively, conditioning on intervening at least once between date t and $t + 5$. Column (6) shows the sample period when FXI data is available.

Table A3: Sample Firms

Country	Total	Dollar Debt	Country	Total	Dollar Debt
Argentina	34	25	Colombia	22	9
Australia	1190	126	Hong Kong	480	42
Brazil	68	21	Japan	2216	4
Chile	3	1	Mexico	48	33
			Total	4060	261

Note: The table shows the number of all firms and firms with Dollar debt in each country. For firms with Dollar debt, the table shows the number of firms which issued Dollar debt at least once during the sample period.

Table A4: Stock Price: Effect of Intervention

Dependent variable:	$\Delta \text{Stock Price}_{i(c),t}$			
	Dollar Debt	No Dollar Debt	Both	
	(1)	(2)	(3)	(4)
FFR Shock _t	-0.647*** (0.116)	-0.093** (0.045)	-0.096** (0.045)	
FFR Shock _t × Intervention _{c,t}	0.449*** (0.130)	-0.035 (0.082)	-0.042 (0.079)	
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}			-0.310*** (0.083)	-0.259*** (0.070)
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)}			0.324*** (0.093)	0.232*** (0.067)
R^2	0.091	0.033	0.033	0.086
N	3,206	113,534	116,754	116,754
Firm FE	✓	✓	✓	✓
Country × Date FE				✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Intervention_{c,t} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt_{i(c),y-1(t)} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A5: Stock Price: Interaction (Controlling for Financial Channels)

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _t	-0.097** (0.045)	-0.033 (0.034)	-0.040 (0.043)	-0.095** (0.046)	-0.096** (0.045)	0.005 (0.036)
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}	-0.314*** (0.087)	-0.255*** (0.083)	-0.232** (0.093)	-0.316*** (0.077)	-0.284*** (0.105)	-0.202** (0.098)
R ²	0.031	0.044	0.041	0.034	0.032	0.053
N	105,114	105,114	97,418	105,114	105,114	97,418
VIX		✓				✓
Currency Risk Premium			✓			✓
Foreign Yield				✓		✓
Capital Flows					✓	✓
Firm FE	✓	✓	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Dollar Debt_{i(c),y-1(t)} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. In column (1), we do not control for the VIX index, currency risk premium, government bond yield, and equity flows. In column (2), we control for the VIX index. In column (3), we control for the currency risk premium. In column (4), we control for the government bond yield. In column (5), we control for equity flows. In column (6), control for the VIX index, currency risk premium, government bond yield, and equity flows at the same time. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double-clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A6: Stock Price: Maturity of Dollar Debt

Dependent variable:	$\Delta \text{Stock Price}_{i(c),t}$			
	Benchmark	Matures 2Q	Matures 3Q	Matures 1Y
	(1)	(2)	(3)	(4)
FFR Shock _t	-0.103* (0.052)	-0.111** (0.053)	-0.109** (0.053)	-0.109** (0.053)
FFR Shock _t × Intervention _{c,t}	-0.035 (0.094)	-0.026 (0.092)	-0.028 (0.092)	-0.028 (0.093)
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}	-0.310*** (0.103)	-0.552** (0.228)	-0.660*** (0.176)	-0.585*** (0.132)
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)}	0.326** (0.127)	0.540* (0.318)	0.564** (0.279)	0.560*** (0.174)
R^2	0.033	0.033	0.033	0.033
N	116,757	116,757	116,757	116,757
Firm FE	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Dollar Debt_{i(c),y-1(t)} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. In column (1), the dummy takes one for dollar debt regardless of maturity. In column (2), (3), and (4), the dummy takes one if firms have dollar debt and the dollar debt matures two quarters, three quarters, and one year around the FOMC announcement date, respectively. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A7: Stock Price: Intensive and Extensive Margins of Dollar Debt

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$			
	Low \$ Debt	High \$ Debt	All \$ Debt	Continuous \$ Debt
	(1)	(2)	(3)	(4)
FFR Shock _t	-0.097** (0.045)	-0.098** (0.046)	-0.099** (0.046)	-0.102** (0.046)
FFR Shock _t × Intervention _{c,t}	-0.039 (0.079)	-0.040 (0.080)	-0.041 (0.080)	-0.035 (0.080)
FFR Shock _t × Dollar Debt	-0.303*** (0.086)	-0.311*** (0.083)	-0.345*** (0.070)	-0.043*** (0.011)
FFR Shock _t × Intervention _{c,t} × Dollar Debt	0.293*** (0.082)	0.345*** (0.073)	0.468*** (0.145)	0.044*** (0.010)
R^2	0.033	0.033	0.033	0.033
N	116,754	116,754	116,754	116,754
Firm FE	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_t is the Fed funds rate shock in basis points. Intervention_{c,t} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt_{i(c),y-1(t)} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Columns (1) and (2) estimate the specification for firms whose dollar debt over total debt ratio is above the 25th and 50th percentiles, respectively. Column (3) estimates the specification for firms that only issue dollar debt. Column (4) uses a standardized share of dollar debt over total debt. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A8: Stock Price: Other Robustness Checks

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$								
	Mean FXI	p75 FXI	Daily Policy Rate	Zero FFR	Excl. GFC	Excl. ZLB	Market Beta	Int Asset Sales	Stock Denom.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
FFR Shock _{<i>t</i>}	-0.108** (0.052)	-0.116** (0.053)	-0.128** (0.051)	-0.115** (0.053)	-0.111** (0.054)	-0.125* (0.065)	-0.117* (0.065)	-0.095* (0.049)	-0.185** (0.082)
FFR Shock _{<i>t</i>} × Intervention _{<i>c,t</i>}	-0.106 (0.096)	0.110 (0.113)	0.018 (0.126)	-0.029 (0.098)	-0.018 (0.100)	-0.022 (0.097)	0.080 (0.096)	0.003 (0.089)	0.056 (0.130)
FFR Shock _{<i>t</i>} × Dollar Debt _{<i>i(c),y-1(t)</i>}	-0.253*** (0.088)	-0.212*** (0.079)	-0.276*** (0.101)	-0.294*** (0.102)	-0.312*** (0.106)	-0.306*** (0.111)	-0.307*** (0.105)	-0.323*** (0.104)	-0.383*** (0.121)
FFR Shock _{<i>t</i>} × Intervention _{<i>c,t</i>} × Dollar Debt _{<i>i(c),y-1(t)</i>}	0.307** (0.124)	0.212* (0.115)	0.293** (0.129)	0.304** (0.124)	0.337** (0.129)	0.323** (0.135)	0.275** (0.118)	0.331** (0.128)	0.382*** (0.143)
<i>R</i> ²	0.033	0.033	0.034	0.022	0.037	0.036	0.033	0.034	0.038
N	116,757	116,757	109,809	188,394	99,258	85,267	116,507	110,770	116,757
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓	✓

Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_{*t*} is the Fed funds rate shock in basis points. Intervention_{*c,t*} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt_{*i(c),y-1(t)*} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Column (1) defines counter-intervention so that the central banks' average net sales of US dollars between dates t and $t + 5$ are positive when the Fed funds rate increases on date t , and vice versa when the Fed funds rate decreases. Column (2) defines unexpected counter-intervention if the residual from estimating Equation 1 is larger than the 75th percentile. Column (3) controls for the daily monetary policy rate. Column (4) includes FOMC event dates with a zero Fed funds rate shock. Column (5) excludes the Global Financial Crisis. Column (6) excludes the period when the policy rate is below or equal to zero. Column (7) controls for the market beta and its interaction with the Fed funds rate shock. Column (8) controls for one-year lagged international assets over total assets ratio and international sales over total sales ratio. Column (9) denominates the stock price in US dollars. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A9: Exchange Rate: Robustness Checks

Dependent Variable	$\Delta\text{Exchange Rate}_{c,t}$					
	Mean FXI	p75 FXI	Daily Policy Rate	Zero FFR	Excl. GFC	Excl. ZLB
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _t	0.146** (0.049)	0.121** (0.041)	0.274** (0.119)	0.210** (0.084)	0.243*** (0.067)	0.201** (0.067)
Intervention _{c,t}	0.181 (0.168)	-0.047 (0.162)	0.303 (0.205)	0.157** (0.061)	0.217 (0.158)	0.247 (0.152)
FFR Shock _t × Intervention _{c,t}	-0.117** (0.044)	-0.150* (0.076)	-0.363** (0.155)	-0.196** (0.081)	-0.231*** (0.068)	-0.208** (0.070)
R^2	0.074	0.071	0.083	0.064	0.107	0.098
N	836	836	683	1,289	754	790
Country FE	✓	✓	✓	✓	✓	✓

Note: $\Delta\text{Exchange Rate}_{c,t}$ is the change in the log of the exchange rate from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. FFR Shock_t is the Fed funds rate shock in basis points. Intervention_{c,t} is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. We control for the trend and standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Column (1) defines counter-intervention so that the central banks' average net sales of US dollars between dates t and $t + 5$ are positive when the Fed funds rate increases on date t , and vice versa when the Fed funds rate decreases. Column (2) defines unexpected counter-intervention if the residual from estimating Equation 1 is larger than the 75th percentile. Column (3) controls for the daily monetary policy rate. Column (4) includes FOMC event dates with a zero Fed funds rate shock. Column (5) excludes the Global Financial Crisis. Column (6) excludes the period when the policy rate is below or equal to zero. Standard errors are in parentheses. Standard errors are double clustered at the country and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A10: Stock Price: Numbers of Hedged and Unhedged Firms

	(1)	(2)	(3)
	Total firms	Hedged firms	Unhedged firms
		Corr(exchange, stock) > 0	Corr(exchange, stock) < 0
(a) Dollar Debt	261	64	195
(b) No Dollar Debt	3709	2409	1378
(c) Export	501	462	38
(d) No Export	3559	2011	1535

Note: the table shows the number of hedged and unhedged firms. Hedged and unhedged firms are defined as those whose stock price has positive and negative correlation with the exchange rate, respectively. The stock price is in terms of local currency. The exchange rate is defined as the value of dollar in terms of local currency so that higher value of exchange rate implies local depreciation. Rows (a) and (b) are firms with and without dollar debt and (c) and (d) are firms with and without exports. Column (1) is the total number of firms and (2) and (3) are hedged and unhedged firms, respectively.

Table A11: Stock Price: Exclude Hedged Firms

	(1)	(2)	(3)
FFR Shock _{<i>t</i>}	-0.111* (0.064)	-0.104** (0.051)	-0.111* (0.064)
FFR Shock _{<i>t</i>} × Intervention _{<i>c,t</i>}	-0.054 (0.115)	-0.025 (0.093)	-0.040 (0.116)
FFR Shock _{<i>t</i>} × Dollar Debt _{<i>i(c),y-1(t)</i>}	-0.293*** (0.101)	-0.318*** (0.107)	-0.297*** (0.104)
FFR Shock _{<i>t</i>} × Intervention _{<i>c,t</i>} × Dollar Debt _{<i>i(c),y-1(t)</i>}	0.288*** (0.103)	0.333** (0.146)	0.281** (0.127)
R^2	0.032	0.033	0.032
N	49,665	101,768	48,912
Excluding Hedged Firms	✓		✓
Excluding Exporters		✓	✓

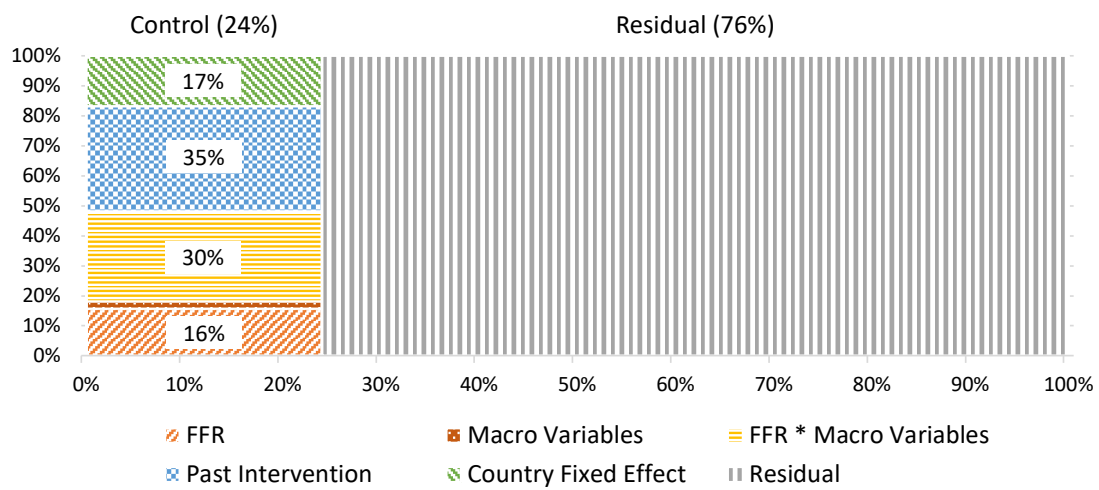
Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock_{*t*} is the Fed funds rate shock in basis points. Dollar Debt_{*i(c),y-1(t)*} is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Column (1) excludes hedged firms, whose stock price has a positive correlation with the exchange rate. Column (2) excludes exporters. Column (3) excludes both hedged firms and exporters. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A12: Stock Price: Cross-Country Heterogeneity

Dependent Variable	$\Delta \text{Stock Price}_{i(c),t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
FFR Shock _t	-0.096** (0.045)	-0.096** (0.045)	-0.095** (0.046)	-0.124** (0.060)	-0.119* (0.061)	-0.053 (0.041)
FFR Shock _t × Intervention _{c,t}	-0.042 (0.079)	-0.049 (0.079)	-0.040 (0.073)	0.031 (0.116)	0.022 (0.119)	-0.087 (0.073)
FFR Shock _t × Dollar Debt _{i(c),y-1(t)}	-0.310*** (0.083)	-0.332*** (0.093)	-0.307*** (0.083)	-0.275*** (0.069)	-0.254*** (0.087)	-0.254*** (0.079)
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)}	0.324*** (0.093)	0.304*** (0.101)	0.343*** (0.098)	0.298*** (0.090)	0.288** (0.110)	0.213*** (0.077)
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)} × EME _c		-0.227 (0.200)				
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)} × Capital Openness _{c,y-1(t)} ^{de jure}			0.026 (0.079)			
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)} × Capital Openness _{c,y-1(t)} ^{de facto}				0.348 (0.392)		
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)} × Trade Openness _{c,y-1(t)}					-0.418 (0.824)	
FFR Shock _t × Intervention _{c,t} × Dollar Debt _{i(c),y-1(t)} × GDP per capita _{c,y-1(t)}						-0.017 (0.059)
R ²	0.033	0.033	0.034	0.034	0.034	0.034
N	116,754	116,754	116,754	116,754	116,754	116,754
Firm FE	✓	✓	✓	✓	✓	✓

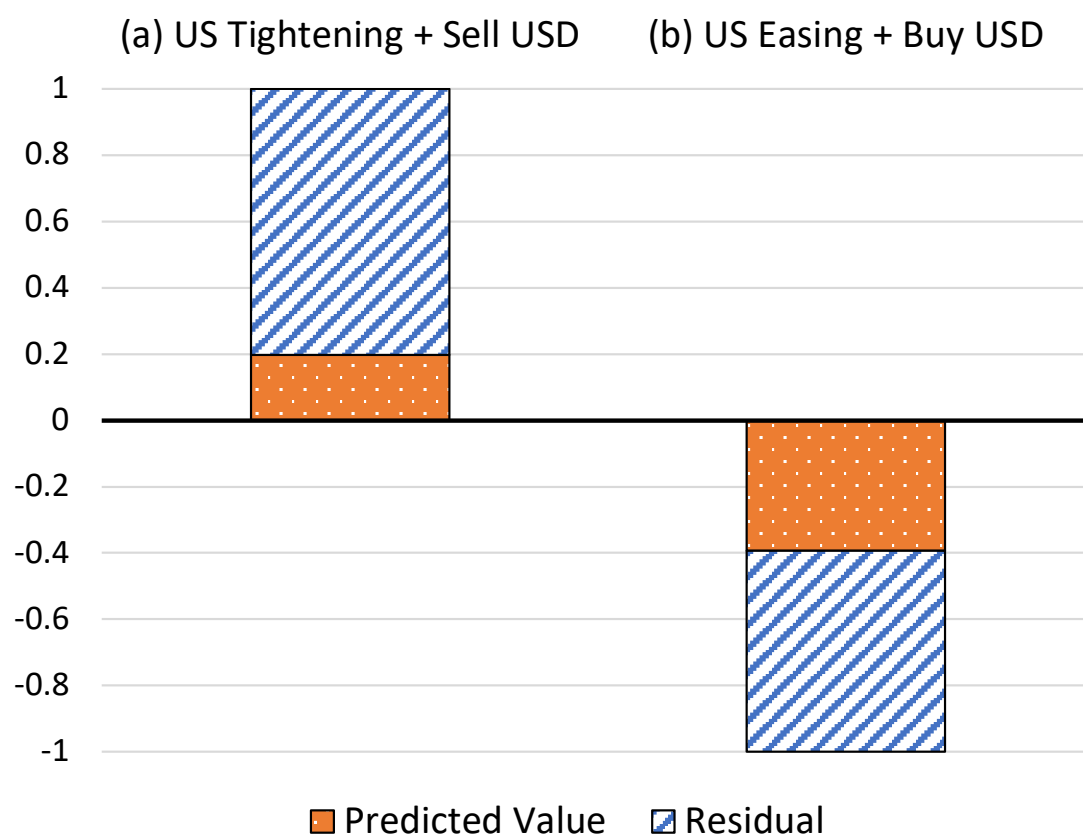
Note: $\Delta \text{Stock Price}_{i(c),t}$ is the change in the log of firm-level stock prices from date $t - 1$ to $t + 1$, where t is the FOMC announcement date. FFR Shock $_t$ is the Fed funds rate shock in basis points. Intervention $_{c,t}$ is an indicator that takes one if there is an unexpected counter-intervention and zero otherwise. Dollar Debt $_{i(c),y-1(t)}$ is an indicator that takes one for firms with dollar debt in the previous year and zero otherwise. Column (2) uses an indicator variable, *EME*, which takes one for emerging economies and zero for advanced economies. Column (3) uses the *KAOPEN* index developed by [Chinn and Ito \(2006\)](#) as a de jure measure of capital account openness. Column (4) uses the sum of gross external assets and liabilities measured as a percentage ratio of GDP as a de facto measure of capital account openness ([Lane and Milesi-Ferretti 2018](#)). Column (5) uses the trade openness measure, defined as the sum of exports and imports measured as a percentage of GDP. Column (6) exploits heterogeneity across GDP per capita. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Figure A1: Variance Decomposition for Counter-Intervention



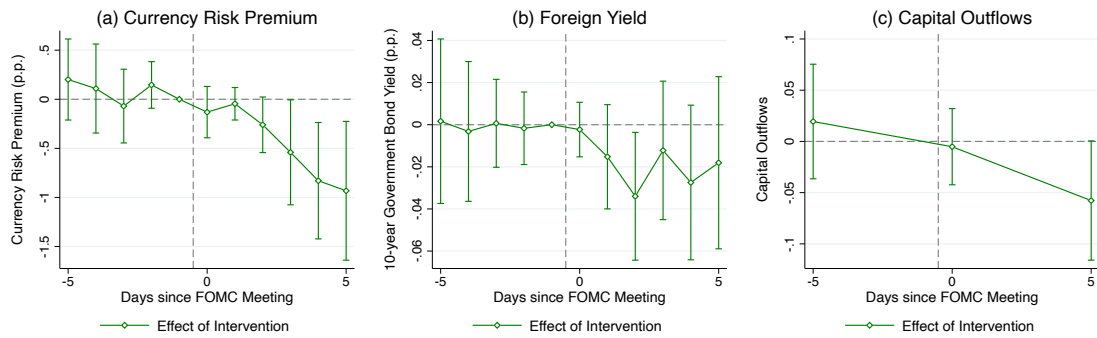
Note: The figure shows the results for variance decomposition for Equation (1). We control for the Fed funds rate shock, one-month lagged policy rate, one-year lagged GDP, CPI inflation rate, trade balance over GDP ratio, unemployment rate, their interaction with the Fed funds rate shock, and FXIs before the FOMC event dates. We include country fixed effect.

Figure A2: Example for Estimating Policy Rule



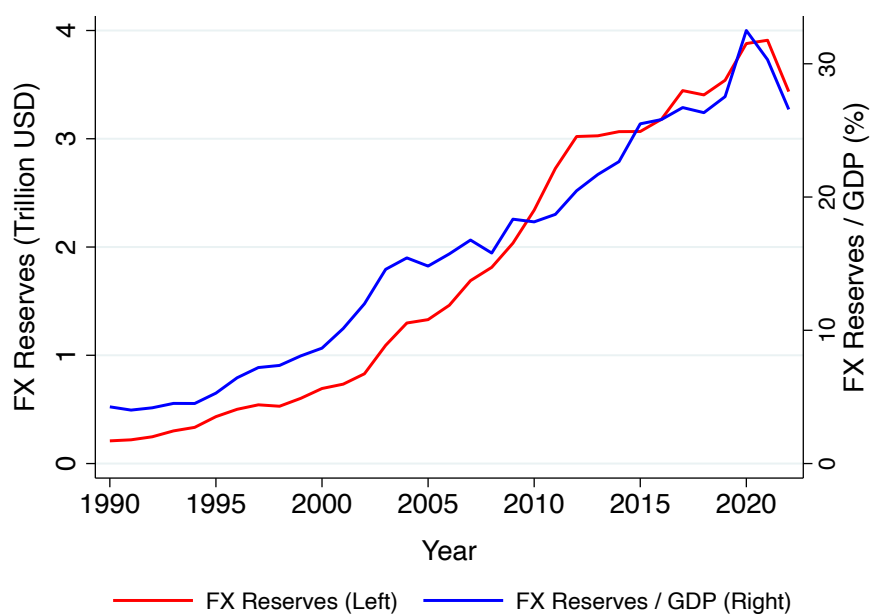
Note: The figure shows the result for the variance decomposition for Equation (1). Column (a) shows the result for a US monetary tightening and US dollar sales intervention by the Reserve Bank of Australia on November 15, 2000. Column (b) shows the result for a US monetary easing and US dollar purchase intervention by the Central Bank of Argentina on March 22, 2005.

Figure A3: Different Transmission Channels of FXI: Difference-in-Differences



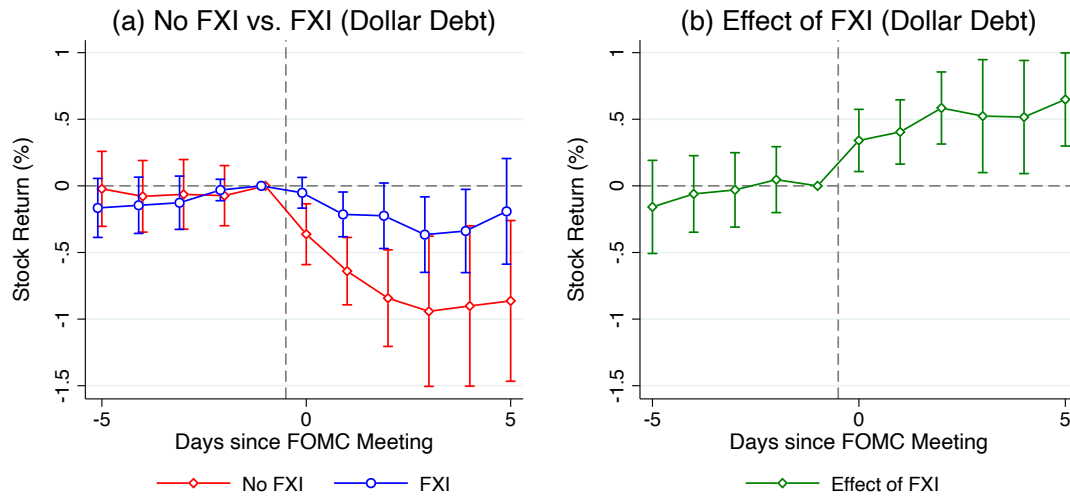
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on the currency risk premium, foreign bond yield, and equity capital flows. In panel (a), the dependent variable is the change in currency risk premium. A positive value implies a higher excess return on the foreign currency over the US dollar. In panel (b), the dependent variable is the change in foreign 10-year government bond yields. In panel (c), the dependent variable is the equity capital flows. A positive value implies a capital outflow from foreign countries. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are double clustered at the country and date level (panel a) and clustered at the date level (panels b and c). Note that, given low clusters due to data availability, we cannot double cluster in panels (c) and (d). The confidence interval is 90%.

Figure A4: Foreign Exchange Reserves in Sample Countries



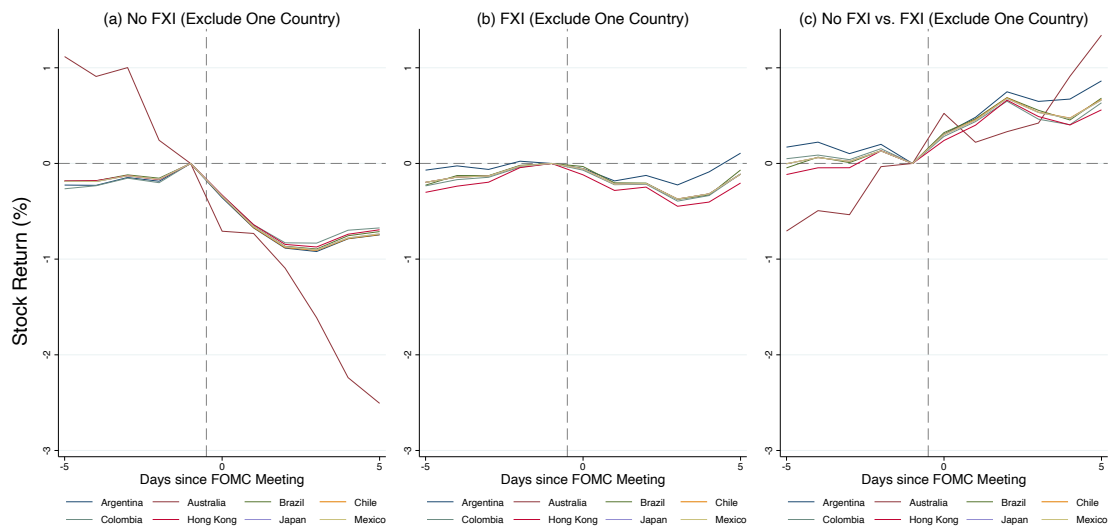
Note: The figure reports the volume of foreign exchange (FX) reserves (left) and the FX reserves over GDP ratio in our sample of 13 countries. For the FX reserves over GDP ratio, we took the ratio of total FX reserves over total GDP in our sample countries. Source: IMF International Financial Statistics.

Figure A5: Stock Price: Weight by the Number of Firms with Dollar Debt



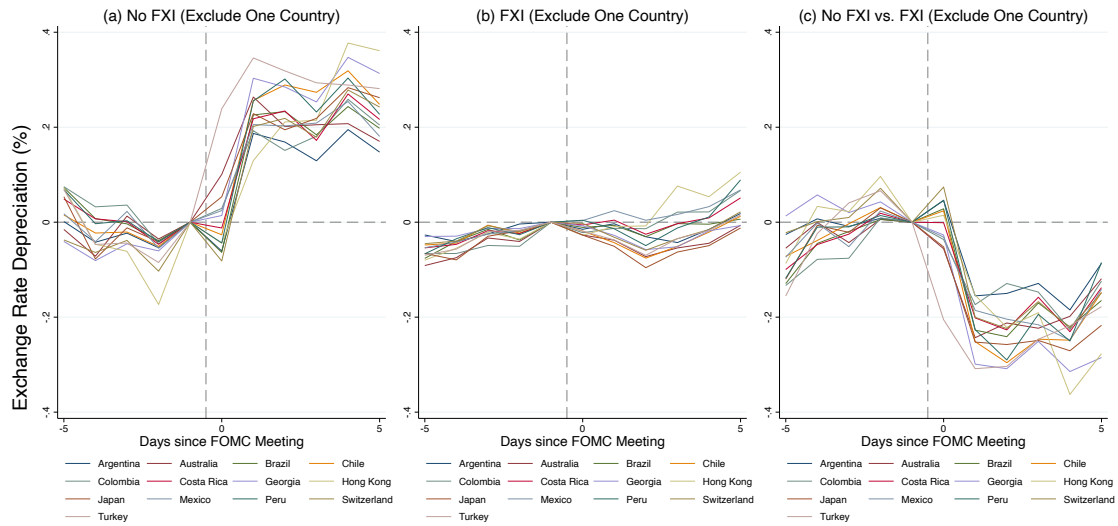
Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. See Equations 2 and 4 for the exact specifications. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. We weight the regression by $1/n_{i(c)}$, where $n_{i(c)}$ is the number of firms with dollar debt in country c where firm i belongs. The confidence interval is 90%.

Figure A6: Stock Price: Exclude Each Country from Sample



Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. Each line shows the point estimates after excluding each country when estimating Equations 2 and 4. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock.

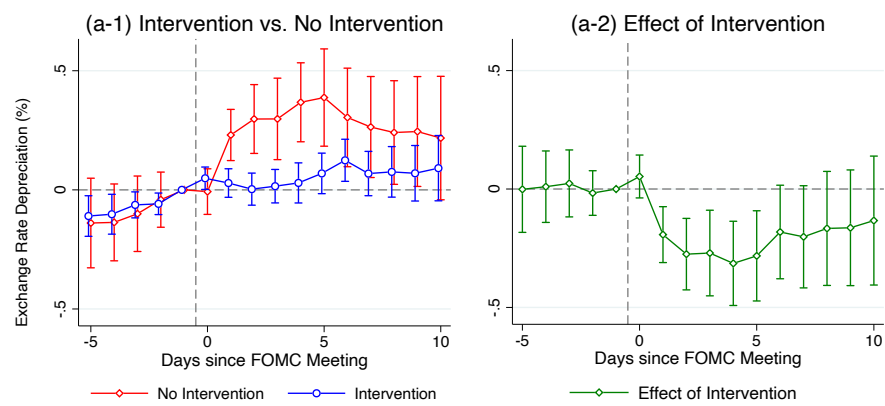
Figure A7: Exchange Rate: Exclude Each Country from Sample



Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on the exchange rate. Each line shows the point estimates after excluding each country when estimating Equations 2 and 4. The exchange rate is defined as the value of the US dollar in terms of local currency, and a higher value implies a depreciation of the local currency. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock.

Figure A8: Different Transmission Channels of FXI (10-day Window)

(a) Exchange Rate



(b) Currency Risk Premium

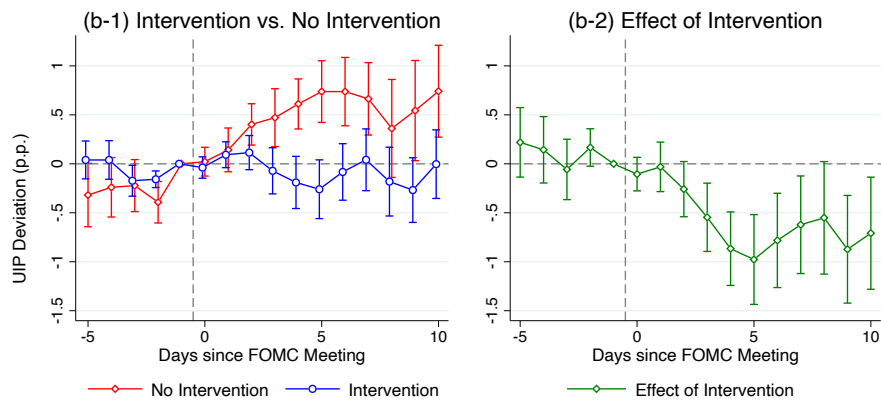
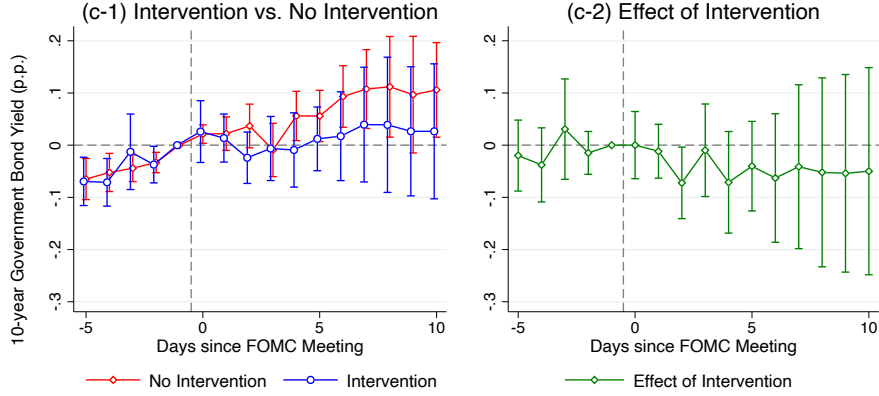
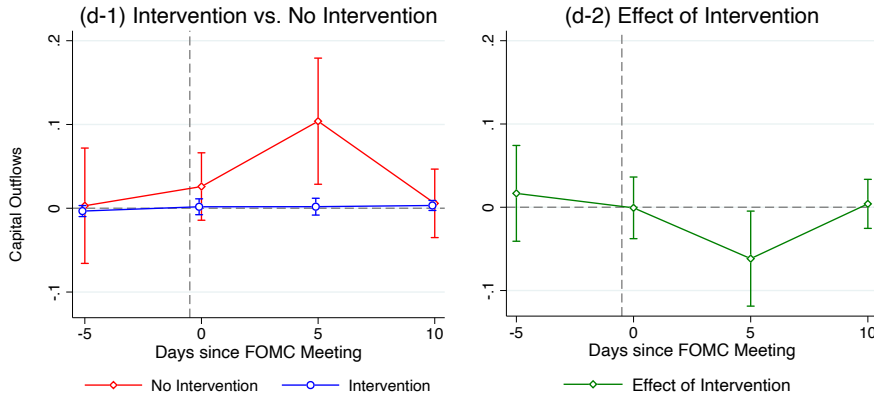


Figure A8: Different Transmission Channels of FXI (10-day Window, Continued)

(c) Foreign Yield

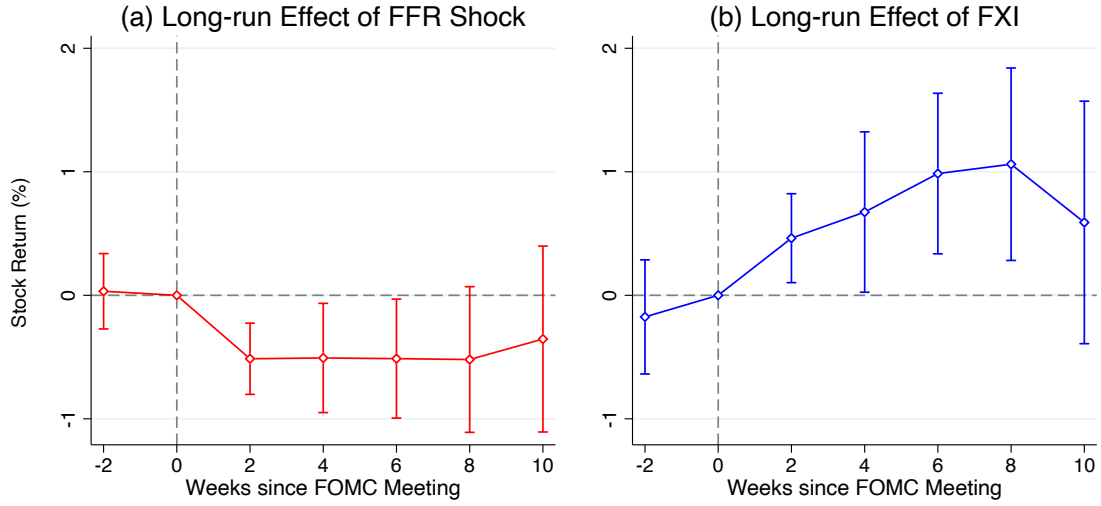


(d) Capital Outflows



Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on the exchange rate, currency risk premium, foreign bond yield, and capital flows. We regress Equation 4 over h days before and after the FOMC announcement, where $h \in [-5, 10]$. In panel (a), the dependent variable is the change in the exchange rate, defined as the value of the US dollar in terms of foreign currency, and a higher value implies a depreciation of the foreign currency. In panel (b), the dependent variable is the change in currency risk premium. A positive value implies a higher excess return on the foreign currency over the US dollar. In panel (c), the dependent variable is the change in foreign 10-year government bond yields. In panel (d), the dependent variable is the equity capital flows. A positive value implies a capital outflow from foreign countries. We control for the standard deviation of the exchange rate before the FOMC announcement date, FXIs before the FOMC announcement date, one-month lagged policy rate, one-year lagged GDP, inflation, trade balance over GDP ratio, unemployment rate, and their interaction with the Fed funds rate shock. Standard errors are clustered at the date level. Note that given low clusters due to data availability, we cannot double cluster at the country and date level. The confidence interval is 90%.

Figure A9: Stock Price: Low-frequency Effect



Note: The figure plots the estimates of the effect of the Fed funds rate shock and FXIs on stock prices. We regress Equation 5 over h weeks before and after the FOMC announcement, where $h \in [-2, 10]$. Panels (a) and (b) plot the coefficients on $FFR_t \times USD_{i(c), y-1(t)}$ and $FFR_t \times USD_{i(c), y-1(t)} \times FXI_{c,t}$, respectively. We control for one-year lagged export intensity, total assets, liquidity-to-asset ratio, firm age (firm-level), one-year lagged import content of production (industry-level), and their interaction with the Fed funds rate shock. We include firm fixed effect and country times date fixed effect. Standard errors are in parentheses. Standard errors are double clustered at the firm and date level. The confidence interval is 90%.